

Atlas-Indexed Reversible Evidence-Fibered Geodesic π PLN

Local Bayesian Charts, Higher Proof Geometry, Revision, and Forgetting as Holonomy

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Abstract

Geodesic inference control treats reasoning as a boundary-value problem: among paths from premise distributions to goal distributions, prefer a least-effort corridor guided by the product of a forward reachability factor and a backward goal-usefulness factor. Paraconsistent Geodesic Modal Homotopy Type Theory (PGM–HoTT) treats identity proofs as graded paths with separate positive and negative evidence channels. The information-geometric theory of energy-based continual learning decomposes forgetting into metric anisotropy and connection curvature, and identifies finite order-dependent forgetting with holonomy around a task loop. This paper integrates these ideas into an atlas-indexed, reversible, directed, evidence-fibered formulation of π PLN.

The persistent cognitive state is not one posterior over one ambient sample space. It is a context-indexed, provenance-rich, paraconsistent evidence metagraph. Ordinary Bayesian or ν PLN calculations occur inside temporary local charts, whose assumptions, priors, factorization choices, and concept boundaries are explicit. Charts export empirical positive and negative evidence back to the persistent layer; gluing charts into a larger probabilistic model is a theorem to be earned, not a global axiom.

Mathematically, the architecture consists of an atlas-indexed family of directed higher proof spaces and evidence fibrations. Their Grothendieck construction contains both within-context inference and guarded context translation. Local Bayesian charts are modeled as reversible evidence/probability lenses: in the beta–Bernoulli case, subtracting the chart prior on export gives an exact round trip and proves that repeated chart construction cannot manufacture evidence. Cross-context de-duplication is permitted only after translation to a covering context. Probabilistic gluing is expressed as a homotopy-descent type; geodesic contractibility licenses an essentially unique global chart, while emptiness records genuine non-gluability. A quantitative descent obstruction is kept separate from curvature.

The paper proves an atlas-level local-to-global coherence theorem whose error is the sum of defects of proof-rewrite cells, inference–translation naturality cells, context-composition cells, chart round trips, and revision cells. In the smooth reversible limit, proof curvature, context curvature, and mixed curvature are the bidegree components of one total evidence connection, and endpoint order dependence is exactly its holonomy. The full diagnostic therefore has three conceptually distinct parts: control-path multiplicity versus unique evidence, reversible holonomy, and probabilistic descent obstruction. Revision becomes a first-class higher morphism, with residuated fit, preservation, weakness, provenance correction, and a reversible complement.

We derive a hybrid geodesic controller that runs Schrodinger bridges inside local Bayesian islands and uses quantale-valued costs for context changes, guarded translations, and revisions. We give curvature-, novelty-, and descent-aware BGC scores; covering-context splice tests; Dykstra-style chart and evidence residuals; and concrete MORK records for contexts, charts,

transitions, descent cells, and reversible revisions. The resulting architecture searches a higher atlas of proofs while transporting an auditable, non-duplicating, reversible evidence section over it.

Contents

1	Introduction	6
1.1	Contributions and theorem status	7
2	Background I: PLN and geodesic inference control	8
2.1	PLN truth values and evidence	8
2.2	Schrodinger-bridge control	8
2.3	A precision concerning Noether conservation	9
3	Background II: PGM–HoTT, SENF, and TransWeave	9
3.1	Split quantale and graded identity	9
3.2	Why a directed extension is needed	10
3.3	Proof relevance before truncation	10
4	Background III: πPLN as an atlas of local Bayesian charts	10
4.1	Persistent packet semantics	10
4.2	Contexts and chart construction	11
4.3	The packet algebra and the evidence-count quantale	11
4.4	STV and beta projections	12
4.5	Exporting evidence rather than priors	12
4.6	Non-gluability is a real probabilistic phenomenon	12
4.7	Residuated revision	13
4.8	Local Bayesian islands in a strict control graph	13
5	Background IV: forgetting as metric anisotropy and curvature	14
5.1	Mode-resolved metric mismatch	14
5.2	Commutators and connection curvature	14
5.3	Finite holonomy	15
5.4	Causal coding and sparse confusing pairs	15
6	A directed evidence-fibered extension of PGM–HoTT	15
6.1	Grades and judgments	15
6.2	Directed higher proof space	16
6.3	Evidence fibration	16
6.4	Evidence continuity	17
6.5	Evidence-preserving proof identity	17

7	Exact evidence de-duplication and approximate proof coherence	18
7.1	The double-counting functional	18
7.2	Homotopy de-duplication	18
7.3	Approximate de-duplication	18
7.4	Proof squares as local curvature cells	19
7.5	Local-to-global proof-surface theorem	19
8	Control mass, evidence mass, and proof multiplicity	20
8.1	Two measures on proof space	20
8.2	An exact mode-resolved identity	21
8.3	Overcounting and undercounting	21
9	The smooth evidence connection and the PLN curvature decomposition	22
9.1	Reduction hypotheses	22
9.2	Conditional curvature theorem	22
9.3	Directed bigons and finite relative holonomy	23
10	Reversible PLN	23
10.1	Logical versus computational reversibility	23
10.2	Graph dilation of an arbitrary rule	24
10.3	How much ancilla is necessary?	24
10.4	Bregman projections and dual residuals	24
10.5	Dykstra variables as reversible projection ancillas	25
10.6	Exact group-commutator holonomy	25
10.7	Visible forgetting as projected reversible holonomy	26
10.8	Reversible idempotent merge	27
11	Reversible HoTT microphysics and directed PLN macrophysics	27
11.1	The microstate groupoid	27
11.2	Literal proof loops	27
11.3	TransWeave as a pseudofunctor on proof space	28
12	Atlas-indexed dE–PGM–HoTT for πPLN	28
12.1	The context category and guarded translations	28
12.2	Doubly indexed evidence	29
12.3	Charts as a contravariant prestack	29
12.4	Local Bayesian charts as reversible lenses	29
12.5	Context-guarded evidence equivalence	30
12.6	Homotopy descent for probabilistic charts	31
12.7	A quantitative descent obstruction	32
12.8	Three kinds of local coherence cell	32

12.9 Atlas local-to-global coherence	32
12.10 Smooth total curvature and exact total holonomy	33
12.11 Holonomy and descent obstruction are independent	33
12.12 Multiplicity, holonomy, and descent	34
12.13 Revision as a first-class higher morphism	34
13 Hybrid geodesic control on the reversible πPLN atlas	35
13.1 Local bridge actions, atlas moves, and fiber constraints	35
13.2 Curvature-, revision-, and evidence-aware BGC	35
13.3 Covering-context and holonomy-aware splicing	36
13.4 Dykstra-PLN	37
14 MORK realization	37
14.1 Rewrite cells and proof classes	37
14.2 Reversible rule applications	38
14.3 Context, chart, descent, and revision records	38
14.4 Local curvature probes	39
14.5 Atlas frontier expansion	40
15 Worked examples	41
15.1 Repeated chart construction does not create evidence	41
15.2 Concept boundaries and covering contexts	41
15.3 A non-gluable chart triangle	42
15.4 Mixed curvature from infer–translate order	42
15.5 Two proof paths, one causal innovation	42
15.6 Noncommuting projection rules	43
15.7 A reversible rule-loop probe	43
16 Empirical predictions and evaluation program	43
16.1 Prior-cycling and chart-roundtrip tests	43
16.2 Holonomy versus non-gluability	44
16.3 Translation naturality predicts ontology errors	44
16.4 Reversible corrective revision	44
16.5 Path multiplicity versus unique evidence	44
16.6 Local commutators predict global path dependence	44
16.7 Exact provenance removes inflation but not curvature	45
16.8 Reversible residuals make forgetting recoverable	45
16.9 Curvature-aware geodesic scheduling	45
16.10 Storage-efficiency frontier	45
17 Limitations and open problems	45

17.1	Choosing the context site and covers	45
17.2	Model-class dependence of descent obstruction	45
17.3	Partial translations and untranslated residue	46
17.4	Priors, hyperpriors, and chart identity	46
17.5	Formal type theory	46
17.6	Causal identity of evidence tokens	46
17.7	Smooth reduction validity	46
17.8	Approximate sketches	46
17.9	Nonabelian proof holonomy	46
17.10	Privacy and reversibility	47
18	Conclusion	47
A	Notation summary	48
B	A compact axiom package for implementation	48

1 Introduction

A large uncertain-reasoning system must solve at least four problems at once.

First, it must decide which inference to perform. Geodesic inference control addresses this as a least-effort boundary-value problem. A good intermediate state is both reachable from the premises and useful toward the goal; in Schrodinger-bridge language these demands are represented by forward and backward factors f and g , with bridge mass proportional to fg [1, 9].

Second, it must decide when two derivations carry the same evidence. A raw proof graph counts syntactically distinct paths. But paraphrases, role permutations, rule reorderings, and repeated projections may all transport the same primitive observations. PGM-HoTT supplies a language for identities between proofs and for graded positive and negative identity evidence [2, 10]. An idempotent provenance fiber then turns those higher identities into an exact de-duplication rule.

Third, it must control order dependence. The coupling-dilemma theory separates a metric term, caused by weighting information modes incorrectly, from a curvature term, caused by noncommuting transport between task or solution manifolds. At finite scale the latter is holonomy [4]. Causal coding adds the structural observation that good causal modularity makes most update pairs commute or nearly commute, so global path dependence is controlled by a sparse graph of genuinely confusing pairs [3].

Fourth, it must reason without pretending that every cognitive context is a marginal of one master probability space. The π PLN formulation treats persistent knowledge as context-indexed positive and negative empirical evidence. Local Bayesian charts are created when a calculation needs ordinary probabilistic coherence, and their outputs are projected back to empirical evidence. Priors belong to charts, not to the durable evidence store. Some chart families glue; others provably do not. Revision therefore acts on evidence packets, provenance, translations, valuations, and ontologies, not only on one global posterior [5].

These requirements lead naturally to an atlas-indexed architecture. Instead of postulating one ambient proof base, associate each context with a directed higher proof space and an evidence fibration, and combine them into a total context-proof structure:

$$\boxed{\begin{array}{c} \text{one proof base with one evidence fibration} \\ \Downarrow \\ \text{a context-indexed atlas of proof bases and evidence fibrations} \end{array}} \quad (1.1)$$

Locally, the de-duplication, proof-surface, reversible-dilation, and holonomy results apply inside each context. Globally, two additional questions arise. Does inference commute with context translation? And do the local probability charts admit any coherent global gluing?

The central synthesis is:

$$\boxed{\begin{array}{c} \text{control mass lives on raw context-proof paths,} \\ \text{evidence mass lives on context-guarded provenance components,} \\ \text{double counting is a mismatch between these two masses,} \\ \text{contextual forgetting is holonomy around proof-and-context loops,} \\ \text{non-gluability is absence of a coherent global chart section.} \end{array}} \quad (1.2)$$

Holonomy and non-gluability must not be conflated. Curvature measures path-dependent transport when transports exist. Descent obstruction says that no global probabilistic object realizes all local charts, even when pairwise overlaps look compatible.

A reversible implementation makes the higher geometry literal. A PLN implication need not be logically reversible, but a particular rule execution can be computationally reversible if it carries the information the visible step would erase: overwritten state, provenance overlap, chart assumptions, prior parameters, Bregman/Dykstra residuals, context labels, translation complements, and random seeds. The visible directed system is then a forgetting projection of an invertible microstate dynamics.

1.1 Contributions and theorem status

The manuscript distinguishes exact discrete statements, exact chart special cases, conditional smooth results, and design proposals.

Result	Status	Content
Evidence de-duplication	Exact	Idempotent evidence transport prevents confidence growth from multiple representatives of one evidential proof class.
Proof-surface bound	Exact	The discrepancy between parallel derivations is bounded by the sum of local proof-rewrite defects.
Beta chart round trip	Exact	Importing empirical counts into a beta chart and exporting after subtracting the local prior is the identity; repeated chart construction cannot duplicate prior mass.
Context-guarded de-duplication	Exact	Cross-context paths merge idempotently only after evidence-preserving transport into a covering context.
Probabilistic non-gluability	Exact example	Three pairwise unbiased anti-correlated charts can agree on all one-variable overlaps and still admit no global joint distribution.
Descent formulation	Semantic theorem schema	A homotopy-descent type represents coherent chart gluings; geodesic contractibility gives an essentially unique global chart.
Atlas proof-surface bound	Exact	Proof, translation, chart, context-composition, and revision defects add along a finite higher surface.
Reversible rule and revision dilation	Exact	Every deterministic visible update is the projection of an injective graph lift; Bregman projections use the dual residual as a geometrically meaningful ancilla.
Total holonomy identity	Exact under invertibility and metric compatibility	The difference between two complete context-proof paths is displacement by their relative group-valued holonomy.
Metric/curvature transfer to PLN	Conditional	Under explicit projection, Fisher, and operating-point hypotheses, the neural forgetting decomposition becomes multiplicity plus proof/context/mixed curvature.

Multiplicity–holonomy– descent diagnostic	Definition and conditional de- composition	Keeps wrong evidence multiplicity, path- dependent transport, and failure of global proba- bilistic gluing conceptually separate.
Hybrid atlas BGC and MORK realization	Design proposal	Local Schrodinger bridges, quantale-valued tran- sitions, revision moves, descent-aware splicing, records, and probes.

The exact atlas results require no global probability distribution, differentiable manifold, or neural energy. The Fisher–Rao formulas are a special smooth realization, not a foundational assumption of π PLN.

2 Background I: PLN and geodesic inference control

2.1 PLN truth values and evidence

A standard PLN truth value may be represented by a strength-confidence pair (s, c) . A convenient evidence parameterization uses positive and negative counts (α, β) . With a confidence scale $k_0 > 0$,

$$w = \alpha + \beta, \quad s = \frac{\alpha}{\alpha + \beta}, \quad c = \frac{w}{w + k_0}. \quad (2.1)$$

Conversely, for $c < 1$,

$$w = k_0 \frac{c}{1 - c}, \quad \alpha = sw, \quad \beta = (1 - s)w. \quad (2.2)$$

The scalar w is not sufficient for sound evidence accounting. Two capsules can each have weight w while being supported by exactly the same primitive observation. Confidence fusion therefore requires provenance, or at least an overlap estimator over provenance.

The local-first MORK design in geodesic inference control associates immutable evidence tokens, sketches, capsule nodes, and rule-application edges with Atomspace expressions. Merge is intended to be associative, commutative, idempotent, and overlap-aware [1]. In the exact model developed below, a capsule carries finite sets of primitive token identifiers. Approximate sketches are treated as a controlled implementation error.

2.2 Schrodinger-bridge control

Let X be a state space, $t \in [t_0, t_1]$, $\rho(x, t)$ a probability density, $v(x, t)$ a velocity field, and $p(x) > 0$ a reference prior. A representative Schrodinger-bridge action is

$$\mathcal{A}_{\text{SB}}[\rho, v] = \int_{t_0}^{t_1} \int_X \rho(x, t) \left[\frac{1}{2} \|v(x, t)\|^2 + \varepsilon \log \frac{\rho(x, t)}{p(x)} \right] dx dt, \quad (2.3)$$

subject to

$$\partial_t \rho + \nabla \cdot (\rho v) = 0, \quad \rho(\cdot, t_0) = \rho_0, \quad \rho(\cdot, t_1) = \rho_1. \quad (2.4)$$

For a suitable reference transition kernel, the bridge has forward and backward factors f and g such that

$$\rho(x, t) = \frac{f(x, t)g(x, t)}{\int_X f(\xi, t)g(\xi, t)d\xi}. \quad (2.5)$$

The control interpretation is immediate. The forward factor measures compatibility with the past boundary; the backward factor measures compatibility with the future boundary. A practical bidirectional geodesic chaining (BGC) heuristic scores a candidate rule step r by

$$\text{Score}_0(r) = \Delta(\log f + \log g) - \lambda \Delta C_r, \quad (2.6)$$

where C_r is local inference effort. Exact f and g may be replaced by factor-graph messages, Monte Carlo meet probabilities, or learned value functions [1].

The crucial point for this paper is that ρ is a *control distribution over paths or partial proof states*. It is not automatically a measure of independent evidence. Twenty cheap derivations can deserve more search attention than one expensive derivation while still carrying only one primitive observation.

2.3 A precision concerning Noether conservation

Time-translation invariance of an action such as (2.3) yields a Hamiltonian or effort-like conserved quantity under the usual regularity hypotheses. It does not, by itself, imply that a primitive evidence token cannot be copied down two branches and later counted twice.

We therefore distinguish:

- (i) *control-energy conservation*, arising from symmetry of the path action;
- (ii) *evidence continuity*, saying that pure inference does not mint new primitive observations;
- (iii) *idempotent fusion*, saying that repeated references to one token contribute once;
- (iv) *flat or curvature-controlled transport*, governing order dependence.

The first is variational. The second and third are algebraic/provenance constraints. The fourth is geometric. Conflating them obscures which intervention is needed when inference goes wrong.

3 Background II: PGM–HoTT, SENF, and TransWeave

PGM–HoTT extends homotopy type theory in three directions [2]. First, identity has separate positive and negative evidence channels. Second, paths have grades in a logic quantale, making shortest or Pareto-minimal proofs first-class. Third, modalities can forget grades or enforce bounded consistency.

3.1 Split quantale and graded identity

Let Q^+ and Q^- be commutative unital quantales. Define

$$V = Q^+ \times (Q^-)^{\text{op}}. \quad (3.1)$$

The reversal in the negative component represents the fact that increasing counter-evidence should weaken positive identity rather than strengthen it. For $x, y : A$ and budgets $\tau^+ \in Q^+$ and $\tau^- \in Q^-$, PGM–HoTT uses graded families

$$\text{Id}_{A, \leq \tau^+}^+(x, y), \quad \text{Id}_{A, \leq \tau^-}^-(x, y). \quad (3.2)$$

Positive and negative identity evidence may coexist without explosion. A left-exact forgetting modality discards grades and returns ordinary HoTT identity; a consistency modality can discharge sufficiently bounded negative evidence.

The theory is modeled by V -enriched infinity-groupoids, supports graded transport, and includes a guarded form of univalence. It underlies a Semantic Elegant Normal Form (SENF) and TransWeave mappings that align semantic frames, roles, entities, times, and regions while preserving graded identity information.

3.2 Why a directed extension is needed

An identity proof in HoTT is invertible. A PLN deduction is generally not. From premises A and $A \Rightarrow B$, deriving B does not produce a deductive inverse from B to the premises. Abduction is a different, uncertain operation.

Consequently, inference rules should not simply be identified with ordinary HoTT paths. The correct separation is:

$$\boxed{\text{semantic identity is invertible, while visible inference is directed.}} \quad (3.3)$$

We will model visible derivations as directed 1-cells in a higher category, while retaining ordinary PGM–HoTT identity in the maximal invertible core. Later, a reversible microstate dilation will show how directed visible rules can nevertheless arise as projections of invertible hidden-state paths.

3.3 Proof relevance before truncation

For a proposition A , let $\text{Der}(\Gamma, A)$ be the full type or higher space of derivations from assumptions Γ . This object must retain syntax, provenance, grades, rewrites, and higher coherence. Mere provability is its propositional truncation,

$$\text{Provable}(A) = \|\text{Der}(\Gamma, A)\|_{-1}. \quad (3.4)$$

If truncation is applied before evidence overlap is resolved, the very distinctions needed for deduplication disappear. The governing principle is therefore

$$\boxed{\text{deduplicate before truncating.}} \quad (3.5)$$

4 Background III: π PLN as an atlas of local Bayesian charts

The π PLN proposal keeps ordinary Bayesian mathematics but changes its scope. The enduring semantics is a context-indexed, paraconsistent evidence metagraph. Probability spaces are temporary local charts used when a task requires them. A global probability space may be constructed when compatibility conditions warrant it, but it is not assumed a priori [5].

4.1 Persistent packet semantics

Let L be a typed metagraph language and $\text{Stmt}(L)$ its statements. A persistent π PLN state has the form

$$K_t : \text{Stmt}(L) \times \text{Ctx} \longrightarrow \mathcal{P}(\text{Packet}) \times Q_{\pm}^{\text{cnt}} \times \mathcal{M}, \quad (4.1)$$

where $\mathcal{M}$ stores provenance, attention, reliability, task traces, assumption fingerprints, and ontology-boundary declarations. For a statement φ in context C , the primary object is a packet family

$$E_t(\varphi, C) = \left\{ e_i = (\Delta_i^+, \Delta_i^-, P_i, a_i, r_i, \dots) \right\}_i. \quad (4.2)$$

The familiar count pair is a cached valuation,

$$(E_{\varphi, C}^+, E_{\varphi, C}^-) = \text{Agg}_C(E_t(\varphi, C)). \quad (4.3)$$

Packets, rather than collapsed counts, remain available for overlap correction, quarantine, retraction, temporal discounting, and ontology relabeling.

4.2 Contexts and chart construction

A π PLN context is a tuple

$$C = (L_C, X_C, E_C, T_C, A_C, W_C, \rho_C), \quad (4.4)$$

where L_C is a local language fragment, X_C a local universe of discourse, E_C selected evidence, T_C the task or query, A_C a temporary assumption package, W_C a weakness or control structure, and ρ_C a relevance/reliability weight.

A chart constructor produces

$$\Gamma_C(K_t, T_C) = (\Omega_C, \mathcal{F}_C, \nu_C, M_C, D_C, \iota_C), \quad (4.5)$$

where $(\Omega_C, \mathcal{F}_C, \nu_C)$ is a local probability space, M_C a local model random variable, D_C a local observation family, and ι_C maps statements to local events. Inside this chart one may use ordinary Bayesian conditioning, sum-product, conjugate updating, or ν PLN rule derivations.

The negative axiom is as important as the positive construction:

It is not required that there exist $(\Omega, \mathcal{F}, \nu)$ with every ν_C as a marginal or restriction.

(4.6)

4.3 The packet algebra and the evidence-count quantale

At the packet level, merge is idempotent on primitive provenance:

$$e \vee e = e. \quad (4.7)$$

At the projected count level, π PLN uses

$$Q_{\pm}^{\text{cnt}} = ([0, \infty]^2, \leq, \vee, \otimes, (0, 0)), \quad (4.8)$$

with componentwise order and join and

$$(n^+, n^-) \otimes (m^+, m^-) = (n^+ + m^+, n^- + m^-). \quad (4.9)$$

This tensor is exact for provenance-disjoint evidence. In general the count valuation v_C is inclusion-exclusion aware:

$$v_C(e_1 \vee e_2) = v_C(e_1) + v_C(e_2) - v_C(e_1 \wedge e_2). \quad (4.10)$$

Thus the additive quantale is a strict algebra for independent sufficient statistics, while the idempotent packet algebra is the more fundamental object for provenance.

4.4 STV and beta projections

Let a chart use prior strength $p_{0,C}$ and virtual count $k_C > 0$. The empirical counts project to a PLN simple truth value

$$s_C = \frac{n^+ + k_C p_{0,C}}{n^+ + n^- + k_C}, \quad c_C = \frac{n^+ + n^-}{n^+ + n^- + k_C}. \quad (4.11)$$

Setting

$$\alpha_{0,C} = k_C p_{0,C}, \quad \beta_{0,C} = k_C (1 - p_{0,C}) \quad (4.12)$$

gives the beta projection

$$\Pi_{\text{Beta}}^C(n^+, n^-) = \text{Beta}(\alpha_{0,C} + n^+, \beta_{0,C} + n^-). \quad (4.13)$$

Theorem 4.1 (Local beta recovery). *Inside a Bernoulli chart containing n^+ positive and n^- negative empirical observations, the projection (4.13) is exactly the beta-Bernoulli posterior determined by the chart prior.*

Proof. The likelihood is proportional to $\theta^{n^+} (1 - \theta)^{n^-}$. Multiplication by the beta prior density $\theta^{\alpha_{0,C}-1} (1 - \theta)^{\beta_{0,C}-1}$ gives the beta posterior with parameters $\alpha_{0,C} + n^+$ and $\beta_{0,C} + n^-$. The empirical counts are sufficient statistics; the prior parameters belong to the chart. $\square$

4.5 Exporting evidence rather than priors

A clean chart export has the form

$$\text{Out}_C(\varphi) = (\hat{n}_C^+(\varphi), \hat{n}_C^-(\varphi), P_C, A_C). \quad (4.14)$$

If a legacy interface emits a Bayesian STV whose strength already includes $(p_{0,C}, k_C)$, the prior must be unwound before persistence. If N_C is the empirical count represented by confidence, then

$$\Delta_C^+(\varphi) = \rho_C((N_C + k_C)s_C - k_C p_{0,C}), \quad (4.15)$$

$$\Delta_C^-(\varphi) = \rho_C(N_C - ((N_C + k_C)s_C - k_C p_{0,C})). \quad (4.16)$$

A negative result or a value outside $[0, N_C]$ is a chart-diagnostic failure, not evidence to be clipped silently.

4.6 Non-gluability is a real probabilistic phenomenon

Pairwise compatible charts need not extend to one global distribution; this is the classical marginal-extension/contextuality phenomenon studied in, for example, [6, 7].

Theorem 4.2 (Non-extendible pairwise marginals). *Let $X, Y, Z \in \{-1, +1\}$. For each pair, define the same unbiased distribution*

$$P(++) = P(--) = \frac{1}{16}, \quad P(+-) = P(-+) = \frac{7}{16}. \quad (4.17)$$

The three pairwise charts agree on every one-variable marginal, but no joint distribution on (X, Y, Z) realizes all three.

Proof. Every one-variable marginal is uniform. Each pair has correlation

$$r = P(X = Y) - P(X \neq Y) = \frac{1}{8} - \frac{7}{8} = -\frac{3}{4}. \quad (4.18)$$

If a joint distribution existed, its correlation matrix

$$R = \begin{pmatrix} 1 & r & r \\ r & 1 & r \\ r & r & 1 \end{pmatrix} \quad (4.19)$$

would be positive semidefinite because $a^\top R a = \mathbb{E}[(a_1 X + a_2 Y + a_3 Z)^2] \geq 0$. Its eigenvalues are $1 - r, 1 - r, 1 + 2r$, and $1 + 2r = -1/2 < 0$. Contradiction. $\square$

This obstruction is stronger than disagreement between two probability estimates. Every local chart is valid and every overlap agrees; the global object does not exist.

4.7 Residuated revision

Let Q_C be a commutative residuated quantale with residual $\Rightarrow_C$, and let P_C be a complete lattice of evaluable propositions or local metagraph patterns. Candidate revised states and evidence packets are projected into this shared lattice by

$$\text{eval}_K : \mathcal{K}_C(H, E) \rightarrow P_C, \quad \text{eval}_E : \mathcal{E}_C \rightarrow P_C. \quad (4.20)$$

For a weakness valuation $w_C : P_C \rightarrow Q_C$, define

$$\text{fit}_C(K; E) = w_C(\text{eval}_E E) \Rightarrow_C w_C(\text{eval}_K K \wedge \text{eval}_E E), \quad (4.21)$$

$$\text{pres}_C(K; H) = w_C(\text{eval}_K H) \Rightarrow_C w_C(\text{eval}_K K \wedge \text{eval}_K H). \quad (4.22)$$

A hard revision maximizes preservation and weakness subject to a fit threshold; a soft revision lexicographically maximizes

$$(\text{fit}_C(K; E), \text{pres}_C(K; H), w_C(\text{eval}_K K), -\text{Cost}(K)). \quad (4.23)$$

This covers empirical addition, overlap discounting, quarantine, retraction, relabeling, context splitting, chart replacement, and ontology revision. Bayesian conditioning is recovered only in probability-valued local special cases.

4.8 Local Bayesian islands in a strict control graph

Inside a chart, probability valuations are generally lax with respect to event intersection: $\nu_C(A \cap B) = \nu_C(A)\nu_C(B)$ only under chart-specific independence assumptions. Outside the chart, exported messages live in strict evidence and control quantales. A practical product label is

$$Q_{\pi\text{PLN}} = Q_{\text{assump}} \times Q_{\pm}^{\text{cnt}} \times Q_{\text{weak}} \times Q_{\text{cost}} \times Q_{\text{prov}}. \quad (4.24)$$

Local probabilistic factor graphs compute their own marginals or sufficient statistics, then export evidence packets. The surrounding graph routes assumptions, evidence, weakness, cost, and provenance; it is not one global probabilistic factor graph.

5 Background IV: forgetting as metric anisotropy and curvature

This section summarizes the part of the coupling-dilemma theory used later. The original setting is an energy-based continual learner, not PLN, but its algebra applies whenever sequential updates and joint assimilation admit the same projection or contraction structure [4].

5.1 Mode-resolved metric mismatch

Let task t have quadratic loss

$$\ell_t(\theta) = \frac{1}{2} \|\theta - \theta_t^*\|_{A_t}^2, \quad A_t \succeq 0, \quad (5.1)$$

with joint information operator

$$M = \sum_t A_t, \quad \lambda_{\min} = \lambda_{\min}(M). \quad (5.2)$$

Sequential training produces mode-dependent recency weights $\bar{w}_{t,k}$, whereas the joint optimum uses precision weights $\bar{b}_{t,k}$. In the commuting mode-resolved model, normalized expected forgetting has the exact form

$$\hat{F}_b = \frac{1}{\lambda_{\min}} \sum_k S_k \sum_t (\bar{w}_{t,k} - \bar{b}_{t,k})^2, \quad (5.3)$$

where S_k are eigenvalues of M . The equation says that even perfectly commuting tasks can forget: sequential learning can assign the wrong multiplicity to evidence in stiff information modes.

5.2 Commutators and connection curvature

For small gradient-flow steps, the Baker–Campbell–Hausdorff expansion gives

$$\prod_t e^{-\tau_t A_t} = \exp \left(- \sum_t \tau_t A_t + \frac{1}{2} \sum_{s < t} \tau_s \tau_t [A_t, A_s] + O(\tau^3) \right). \quad (5.4)$$

The first generator is order-free. All leading order dependence is in the commutators.

When the task Hessians are Fisher metrics of one statistical model evaluated at nearby operating points, the commutator is the contraction of the Fisher–Rao curvature, up to explicit non-exponential-family defects. The resulting decomposition, stated in the coupling-dilemma manuscript as Theorem 26, is schematically

$$\boxed{\hat{F}_b = \underbrace{\frac{1}{\lambda_{\min}} \sum_k S_k \bar{B}_k}_{\text{metric anisotropy}} + c \underbrace{\sum_{s < t} \langle \Xi_{st}, [(\delta_s \wedge \delta_t) \lrcorner R^\perp] \Xi_{st} \rangle}_{\text{connection curvature}} + E_{\text{gen}}}. \quad (5.5)$$

The metric term can be changed by natural-gradient or preconditioning methods. The curvature term is invariant under such a metric-only fix; it requires modularity, causal factorization, or another way of flattening the relevant connection.

5.3 Finite holonomy

The leading-order commutator sum has a finite nonperturbative completion. Let Γ be the loop formed by a sequential task path followed by the reverse joint geodesic, and let $e_{\parallel}$ be the residual direction transported around the loop. Then

$$\mathcal{F}_{\text{order}} = \frac{1}{2} \left\| (\text{Hol}_{\Gamma}(\nabla^{\perp}) - I) e_{\parallel} \right\|_g^2. \quad (5.6)$$

In rank two, if the holonomy angle is Θ ,

$$\mathcal{F}_{\text{order}} = 2 \left\| e_{\parallel} \right\|_g^2 \sin^2 \frac{\Theta}{2}. \quad (5.7)$$

In higher rank, an orthogonal holonomy decomposes into rotation planes with angles Θ_i and projectors P_i :

$$\mathcal{F}_{\text{order}} = 2 \sum_i \sin^2 \frac{\Theta_i}{2} \left\| P_i e_{\parallel} \right\|_g^2. \quad (5.8)$$

Dykstra correction variables resum the residuals omitted by ordinary cyclic projection and reconstruct the joint projection; geometrically, they carry the transport defect whose cycle product is holonomy.

5.4 Causal coding and sparse confusing pairs

Causal coding seeks a factorization in which each context updates only the causal modules it actually influences. In the smooth setting this suppresses cross-module Hessian blocks and Lie brackets. In a metric-enriched categorical setting, let U_c be the actual update, V_c an ideal context-local update, and

$$d(U_c, V_c) \leq \varepsilon_c, \quad \Delta_{cc'} = d(V_c V_{c'}, V_{c'} V_c). \quad (5.9)$$

Then the actual pairwise confusion cost is bounded by

$$\delta_{cc'} = 2(\varepsilon_c + \varepsilon_{c'}) + \Delta_{cc'}. \quad (5.10)$$

An adjacent-swap argument shows that global path dependence is at most a sum over the confusing pairs actually crossed when one order is transformed into another [3]. This discrete local-to-global pattern will be reused directly for proof rewrites.

6 A directed evidence-fibered extension of PGM–HoTT

We now define the core mathematical structure. It will be called dE–PGM–HoTT: directed, evidence-fibered PGM–HoTT. The name describes a semantic extension rather than a completed proof-assistant implementation.

6.1 Grades and judgments

Let

$$\mathbb{G} = Q^+ \times (Q^-)^{\text{op}} \times Q_{\text{eff}} \times Q_{\text{dup}} \times Q_{\text{coh}} \quad (6.1)$$

be a product quantale. The first two components are inherited from PGM–HoTT. The remaining components may record inference effort, an upper bound on duplicate-evidence exposure, and an upper bound on coherence defect.

The core judgments are

$$\Gamma \vdash x : \mathcal{B}, \quad \Gamma \vdash p : x \Rightarrow_g y, \quad \Gamma \vdash \alpha : p \Rightarrow_\omega q, \quad \Gamma \vdash e : \text{Ev}(x). \quad (6.2)$$

Here p is a directed derivation, α is a 2-cell or proof rewrite, and e is evidence in the fiber above x .

Composition is graded:

$$p : x \Rightarrow_g y, \quad q : y \Rightarrow_h z \quad \Longrightarrow \quad q \circ p : x \Rightarrow_{g \otimes h} z. \quad (6.3)$$

Horizontal and vertical composition of 2-cells accumulate coherence grade in the corresponding quantale component.

6.2 Directed higher proof space

Definition 6.1 (Directed proof base). A directed proof base is a $\mathbb{G}$ -enriched complete Segal object $\mathcal{B}$. Its objects are belief or inference states, its directed 1-cells are rule applications and composite derivations, and its higher cells are proof rewrites and their coherences.

The maximal invertible subobject $\text{Core}(\mathcal{B})$ is an infinity-groupoid. We require it to carry the split $V = Q^+ \times (Q^-)^{\text{op}}$ enrichment of PGM–HoTT. Thus semantic identity, TransWeave equivalences, and graded positive/negative identity proofs remain invertible even though ordinary inference arrows are not.

Remark 6.2. A complete Segal object is used because composition should be coherent rather than strictly associative. Readers who do not need this homotopy-level generality may replace $\mathcal{B}$ by a strict 2-category or even a directed graph with specified rewrite squares. All exact metric bounds below already hold in the strict discrete case.

6.3 Evidence fibration

Definition 6.3 (Evidence fibration). An evidence fibration over $\mathcal{B}$ is a cocartesian fibration

$$\pi_{\mathcal{E}} : \mathcal{E} \longrightarrow \mathcal{B} \quad (6.4)$$

whose fiber over x is a metric commutative idempotent evidence algebra $\text{Ev}(x)$. Every derivation $p : x \Rightarrow y$ induces a join-preserving, nonexpansive transport

$$T_p : \text{Ev}(x) \longrightarrow \text{Ev}(y), \quad (6.5)$$

and every 2-cell $\alpha : p \Rightarrow q$ induces a specified comparison

$$\eta_\alpha : T_p \Longrightarrow T_q. \quad (6.6)$$

In the exact token model,

$$\text{Ev}(x) = \mathcal{P}_{\text{fin}}(\mathcal{U}_x^+) \times \mathcal{P}_{\text{fin}}(\mathcal{U}_x^-), \quad (6.7)$$

with join given by componentwise union. Positive and negative evidence are retained separately:

$$(E^+, E^-) \vee (F^+, F^-) = (E^+ \cup F^+, E^- \cup F^-). \quad (6.8)$$

Therefore

$$e \vee e = e. \quad (6.9)$$

A nonnegative weight function on primitive tokens gives valuations ν^+ and ν^- . PLN truth values are computed from these valuations only after union and overlap correction.

6.4 Evidence continuity

Definition 6.4 (Pure inference and observation). A pure inference transport T_p is provenance-noncreating if

$$\text{supp}(T_p e) \subseteq \text{supp}(e). \quad (6.10)$$

An observation morphism may add a fresh primitive token u not already in the support.

Pure inference may change which proposition a token supports, and it may attenuate or transform positive and negative weights according to a truth function. It may not create a second independent copy of the same token. In an implementation, references to one immutable token can fan out through many proof branches; the confidence valuation is still applied to their idempotent union.

6.5 Evidence-preserving proof identity

Definition 6.5 (Evidence equivalence). Two parallel derivations $p, q : x \Rightarrow y$ are strongly evidence-equivalent, written

$$p \simeq_{\mathcal{E}} q, \quad (6.11)$$

if they are connected by a zigzag of 2-cells with zero coherence defect and their induced transports agree as maps $T_p = T_q$.

The evidence components of the derivation space are

$$\pi_0^{\mathcal{E}} \text{Der}(x, y) := \text{Der}(x, y) / \simeq_{\mathcal{E}}. \quad (6.12)$$

This quotient is finer than mere equality of conclusions and coarser than syntactic equality of proofs. Two derivations proving the same proposition with independent observations must remain in different components. Two elaborate derivations that merely repackage one observation should lie in the same component.

Proposition 6.6 (Semantic realization of the core fragment). *Let $\mathcal{B}$ be a $\mathbb{G}$ -enriched complete Segal object whose core is a PGM–HoTT model, and let $\pi_{\mathcal{E}} : \mathcal{E} \rightarrow \mathcal{B}$ be an evidence fibration as above. Then directed composition, graded 2-cell composition, evidence transport, PGM–HoTT identity transport on the core, and the evidence-equivalence relation are all sound in this semantic structure.*

Proof. Directed composition and higher coherence are supplied by the complete Segal structure. Grades compose by the monoidal product in $\mathbb{G}$. Cocartesian lifting supplies T_p , and functoriality gives $T_{q \circ p} \simeq T_q T_p$. The restriction to $\text{Core}(\mathcal{B})$ is an enriched infinity-groupoid, so the original PGM–HoTT identity interpretation applies there. Zero-defect 2-cells compose and invert within the chosen zigzags, and equality of induced transports is transitive, so $\simeq_{\mathcal{E}}$ is an equivalence relation. $\square$

This proposition is deliberately modest. It establishes a semantic package for the fragment used in this paper. A complete syntax with substitution, universes, guarded univalence, and normalization for all new directed/evidence constructs is a separate formalization project.

7 Exact evidence de-duplication and approximate proof coherence

7.1 The double-counting functional

For parallel derivations $P = \{p_1, \dots, p_n\}$ and evidence e , define

$$\mathcal{D}_{\text{dc}}(P; e) = \sum_{i=1}^n \nu(T_{p_i} e) - \nu\left(\bigvee_{i=1}^n T_{p_i} e\right). \quad (7.1)$$

For the exact token model, take $\nu(E) = \sum_{u \in E} w_u$ in either evidence channel.

Proposition 7.1 (Multiplicity formula). *Let m_u be the number of transported capsules $T_{p_i} e$ containing token u . Then*

$$\mathcal{D}_{\text{dc}}(P; e) = \sum_u (m_u - 1)_+ w_u \geq 0. \quad (7.2)$$

The same formula holds separately for positive and negative channels.

Proof. The first term in (7.1) counts token u exactly m_u times. The union in the second term counts it once if $m_u \geq 1$ and zero times otherwise. Subtraction gives $(m_u - 1)_+ w_u$, and summing over tokens proves the claim. $\square$

7.2 Homotopy de-duplication

Theorem 7.2 (Exact evidential homotopy de-duplication). *Suppose evidence merge is idempotent, transports preserve joins, and $p_i \simeq_{\mathcal{E}} p_1$ for every i . Then for every e ,*

$$\bigvee_{i=1}^n T_{p_i} e = T_{p_1} e. \quad (7.3)$$

Therefore adding syntactic representatives of one evidential proof class cannot increase positive or negative evidence weight.

Proof. Evidence equivalence gives $T_{p_i} e = T_{p_1} e$ for every i . By idempotence,

$$T_{p_1} e \vee \dots \vee T_{p_1} e = T_{p_1} e. \quad (7.4)$$

Applying either channel valuation proves the last statement. $\square$

Interpretation. Double counting is not primarily a numerical calibration error. It is a failure to preserve higher proof identity. A flat proof graph sees n paths; the evidence fibration sees one transported provenance class.

7.3 Approximate de-duplication

Assume each evidence fiber has a metric $d_{\mathcal{E}}$ and join is nonexpansive:

$$d_{\mathcal{E}}(a \vee c, b \vee c) \leq d_{\mathcal{E}}(a, b). \quad (7.5)$$

Theorem 7.3 (Approximate de-duplication). *Let $a_i = T_{p_i}e$ and suppose*

$$d_{\mathcal{E}}(a_i, a_1) \leq \omega_i \quad (i \geq 2). \quad (7.6)$$

Then

$$d_{\mathcal{E}}\left(\bigvee_{i=1}^n a_i, a_1\right) \leq \sum_{i=2}^n \omega_i. \quad (7.7)$$

If ν is L_{ν} -Lipschitz, then

$$\left| \nu\left(\bigvee_i a_i\right) - \nu(a_1) \right| \leq L_{\nu} \sum_{i=2}^n \omega_i. \quad (7.8)$$

Proof. Set $b_j = a_1 \vee \dots \vee a_j$. Since $a_1 \vee a_1 = a_1$, nonexpansiveness gives

$$d_{\mathcal{E}}(b_j, b_{j-1}) = d_{\mathcal{E}}(b_{j-1} \vee a_j, b_{j-1} \vee a_1) \leq d_{\mathcal{E}}(a_j, a_1) \leq \omega_j. \quad (7.9)$$

The triangle inequality proves (7.7). The valuation bound follows from Lipschitz continuity. $\square$

This theorem gives an explicit place for sketch error, approximate causal equivalence, or nonexact TransWeave alignment to enter the accounting.

7.4 Proof squares as local curvature cells

Consider two rules r and s that can be applied in either order:

$$\begin{array}{ccc} x & \xrightarrow{r} & x_r \\ s \downarrow & & \downarrow s|r \\ x_s & \xrightarrow{r|s} & x_{rs}. \end{array} \quad (7.10)$$

If the two endpoints are related by a semantic identity $\iota : x_{sr} \simeq x_{rs}$ in the PGM-HoTT core, define the evidence curvature defect of the square by

$$\delta_{\square}(e) = d_{\mathcal{E}}\left(T_{\iota}T_{s|r}T_r(e), T_{r|s}T_s(e)\right). \quad (7.11)$$

A zero-defect 2-cell fills the square coherently. A nonzero defect means that the same semantic endpoint receives different evidence depending on rule order.

7.5 Local-to-global proof-surface theorem

Theorem 7.4 (Proof-surface coherence bound). *Let $p, q : x \Rightarrow y$ be parallel derivations connected by a finite proof surface Σ tiled by local rewrite cells $\sigma_1, \dots, \sigma_m$. Assume evidence transport and contextual pre- and post-composition are nonexpansive. If cell σ_i has defect at most δ_i , then for every $e \in \text{Ev}(x)$,*

$$d_{\mathcal{E}}(T_p e, T_q e) \leq \sum_{i=1}^m \delta_i. \quad (7.12)$$

Consequently,

$$\frac{1}{2}d_{\mathcal{E}}(T_p e, T_q e)^2 \leq \frac{1}{2}\left(\sum_i \delta_i\right)^2. \quad (7.13)$$

Proof. Choose a sequence of derivations

$$p = p_0 \Rightarrow p_1 \Rightarrow \cdots \Rightarrow p_m = q \quad (7.14)$$

that replaces one local cell at each step. Nonexpansiveness of common prefixes and suffixes implies

$$d_{\mathcal{E}}(T_{p_{i-1}}e, T_{p_i}e) \leq \delta_i. \quad (7.15)$$

The triangle inequality gives (7.12); squaring yields the second claim. $\square$

Corollary 7.5 (Sparse confusing-rule bound). *Suppose actual rule transports U_r are within ε_r of ideal local transports V_r , and define*

$$\Delta_{rs} = d(V_s V_r, V_r V_s), \quad \delta_{rs} = 2(\varepsilon_r + \varepsilon_s) + \Delta_{rs}. \quad (7.16)$$

Then the discrepancy between two rule orders is bounded by the sum of δ_{rs} over the local inversions used to transform one order into the other. If only M rule pairs are confusing and $\delta_{rs} \leq \delta$, the bound is $O(M\delta)$ rather than $O(T^2\delta)$.

Proof. The pairwise bound is the triangle-inequality estimate used in the approximate categorical CCL theorem. Apply the proof-surface theorem to the adjacent-swap surface that sorts one order into the other. Each inversion pair is crossed at most once. $\square$

Interpretation. The 2-cells of proof space are not decorative. They are the local carriers of curvature. Causal modularity makes most squares flat because rules with disjoint causal support commute. The global proof-path defect is the accumulated failure of the remaining cells to commute.

8 Control mass, evidence mass, and proof multiplicity

8.1 Two measures on proof space

Let $\rho_{\text{ctl}}(p)$ be the control mass assigned by the fg scheduler to a raw partial derivation p . Let

$$\pi : \text{Der}(x, y) \longrightarrow \pi_0^{\mathcal{E}} \text{Der}(x, y) \quad (8.1)$$

map a derivation to its evidential component C . The pushforward control mass is

$$(\pi_* \rho_{\text{ctl}})(C) = \sum_{p \in C} \rho_{\text{ctl}}(p). \quad (8.2)$$

Separately, let $\mu_{\text{ev}}(C)$ be the unique evidence precision contributed by component C after idempotent provenance fusion.

There is no reason for these measures to coincide. Multiple easy proofs can make a region of proof space highly reachable while carrying one observation. Conversely, a difficult proof can carry rare independent evidence and be underexplored by a cost-only scheduler.

8.2 An exact mode-resolved identity

The next theorem is the proof-space analogue of the mode-resolved forgetting identity.

Theorem 8.1 (Mode-resolved proof-multiplicity identity). *Let $M \succ 0$ have orthonormal eigenvectors u_k and eigenvalues S_k , with $\lambda_{\min} = \min_k S_k$. Let $C = 1, \dots, N$ index unique evidential components. Suppose independent centered innovations $\xi_C \in \mathbb{R}^d$ satisfy*

$$\mathbb{E}[\xi_C] = 0, \quad \mathbb{E}[\xi_C \xi_D^\top] = \sigma^2 \delta_{CD} I. \quad (8.3)$$

Let sequential/geodesic assimilation and joint unique-evidence assimilation be

$$\theta_{\text{seq}} = \mu + \sum_C W_C \xi_C, \quad \theta_{\text{joint}} = \mu + \sum_C B_C \xi_C, \quad (8.4)$$

where W_C and B_C commute with M and have mode entries $w_{C,k}$ and $b_{C,k}$. Then

$$\mathbb{E} \left[\frac{1}{2\lambda_{\min}} \|\theta_{\text{seq}} - \theta_{\text{joint}}\|_M^2 \right] = \frac{\sigma^2}{2\lambda_{\min}} \sum_k S_k \sum_C (w_{C,k} - b_{C,k})^2. \quad (8.5)$$

Proof. Write $D_C = W_C - B_C$. Independence gives

$$\mathbb{E} \left\| \sum_C D_C \xi_C \right\|_M^2 = \sum_C \mathbb{E} \xi_C^\top D_C^\top M D_C \xi_C = \sigma^2 \sum_C \text{tr}(D_C^\top M D_C). \quad (8.6)$$

Since D_C and M are diagonal in the u_k basis,

$$\text{tr}(D_C^\top M D_C) = \sum_k S_k (w_{C,k} - b_{C,k})^2. \quad (8.7)$$

Divide by $2\lambda_{\min}$. □

In a geodesic PLN controller, $w_{C,k}$ is induced by aggregated path occupation or update scheduling, while $b_{C,k}$ is the overlap-corrected precision that a joint assimilation of unique evidence would assign.

8.3 Overcounting and undercounting

Define

$$\Delta_{C,k} = w_{C,k} - b_{C,k}, \quad \Delta_{C,k}^+ = (\Delta_{C,k})_+, \quad \Delta_{C,k}^- = (-\Delta_{C,k})_+. \quad (8.8)$$

Then

$$\Delta_{C,k}^2 = (\Delta_{C,k}^+)^2 + (\Delta_{C,k}^-)^2. \quad (8.9)$$

The positive part is evidence overcounting or repeated reinforcement. The negative part is evidence starvation, displacement, or inferential forgetting.

When $b_{C,k} > 0$, define effective multiplicity

$$m_{C,k} = \frac{w_{C,k}}{b_{C,k}}. \quad (8.10)$$

The contribution is

$$b_{C,k}^2 (m_{C,k} - 1)^2. \quad (8.11)$$

Thus the metric term is not merely “duplicate evidence.” It is the full squared error in how many effective uses the controller gives each unique evidential component.

Definition 8.2 (Proof-space multiplicity error). The mode-weighted control/evidence mismatch is

$$\mathcal{E}_{\text{mult}} = \frac{1}{\lambda_{\min}} \sum_k S_k \sum_C (w_{C,k} - b_{C,k})^2. \quad (8.12)$$

This is the direct proof-space reading of metric anisotropy. A metric or scheduling change can reduce it without changing which proofs are semantically valid.

9 The smooth evidence connection and the PLN curvature decomposition

The discrete results above require no manifold. We now state the hypotheses under which they admit the same smooth interpretation as energy-based continual learning.

9.1 Reduction hypotheses

Assumption 9.1 (Smooth proof-state reduction). There is a smooth manifold Θ of local PLN belief/evidence states, with a positive metric g , such that each rule or evidential component r has a smooth loss ℓ_r and update flow or projection on Θ .

Assumption 9.2 (Projection model). Assimilating rule/evidence component r to local convergence is a Bregman projection onto a solution submanifold $\mathcal{M}_r$. Joint overlap-corrected assimilation is the corresponding joint projection or joint minimum.

Assumption 9.3 (One model, many operating points). The local Hessians $A_r = \nabla^2 \ell_r$ are Fisher or Gauss–Newton metrics of one statistical model evaluated at nearby operating points θ_r , up to bounded defects E_r , curvature drift, and non-autoparallelity.

Assumption 9.4 (Mode identification). The path-control weights and unique-evidence weights in (8.12) coincide with the recency and precision profiles of the projection model.

These are substantial assumptions. They are plausible for a differentiable PLN truth-value layer, a probabilistic circuit realization, or an energy-based neural-symbolic substrate, but they do not hold automatically for every symbolic rule engine.

9.2 Conditional curvature theorem

Theorem 9.5 (Proof-space metric-curvature decomposition). *Under the smooth reduction hypotheses, for sufficiently nearby operating points and small sequential steps, normalized PLN path error satisfies*

$$\boxed{\hat{\mathcal{F}}_{\text{PLN}} = \underbrace{\mathcal{E}_{\text{mult}}}_{\text{control multiplicity versus unique evidence}} + c \underbrace{\sum_{r < s} \left\langle \Xi_{rs}, [(\delta_r \wedge \delta_s) \lrcorner R^{\mathcal{E}, \perp}] \Xi_{rs} \right\rangle}_{\text{evidence-connection curvature}} + E_{\text{approx}}.} \quad (9.1)$$

Here $R^{\mathcal{E}, \perp}$ is the normal curvature of the evidence/solution bundle, and E_{approx} contains Hessian–Fisher mismatch, drift of the third-order tensor, extrinsic curvature, finite-step error, and provenance-sketch error.

Proof. Under the assumptions, the PLN assimilation problem is an instance of the projection model used in the coupling-dilemma theorem. Relabel the task index by the unique evidential component or rule index r . Assumption 4 identifies the first term of that theorem with (8.12). The BCH commutator term is identified, under Assumption 3, with the curvature of the Fisher/evidence connection contracted against the operating-point bivector. The stated defect terms are exactly the violations of the exponential-family, flatness, and exact-provenance hypotheses. Therefore (9.1) is a corollary of the metric-curvature decomposition in (5.5). $\square$

Conceptual reading. The first term is one-dimensional: it concerns how a measure is placed on evidential 1-path components. The second is two-dimensional: it concerns whether evidence transport respects proof 2-cells. The first can be corrected by weighting or preconditioning. The second requires changing causal support, modularity, or connection structure.

9.3 Directed bigons and finite relative holonomy

For noninvertible visible inference, two parallel paths $p, q : x \Rightarrow y$ form a directed bigon rather than a literal loop. The primitive discrepancy is

$$\text{BigonDef}(p, q; e) = d_{\mathcal{E}}(T_p e, T_q e). \quad (9.2)$$

If transport on a residual evidence bundle is invertible and metric-compatible, define relative holonomy

$$H_{p,q} = T_q^{-1} T_p. \quad (9.3)$$

Then

$$\boxed{\mathcal{F}_{\text{order}}(p, q; e) = \frac{1}{2} \|(H_{p,q} - I)e\|^2.} \quad (9.4)$$

When p followed by q^{-1} bounds a proof surface Σ ,

$$H_{p,q} = \text{Sexp} \int_{\Sigma} R^{\mathcal{E}}. \quad (9.5)$$

Equations (5.7) and (5.8) then apply to the evidence residual.

The finite law is stronger than a statement that error is “related to” curvature. It says that the endpoint discrepancy is the actual monodromy of the evidence connection around the proof loop.

10 Reversible PLN

10.1 Logical versus computational reversibility

A reversible implementation of a deduction does not assert the converse implication. It stores enough information to undo the exact state transition. If a visible rule is

$$U : X \longrightarrow Y, \quad (10.1)$$

its inverse at the microstate level means “restore the previous x and all overwritten evidence registers,” not “deduce x from $U(x)$ without those registers.”

10.2 Graph dilation of an arbitrary rule

Theorem 10.1 (Reversible graph dilation). *For every deterministic map $U : X \rightarrow Y$, the graph map*

$$\iota_U : X \longrightarrow \text{Graph}(U) \subseteq Y \times X, \quad \iota_U(x) = (U(x), x), \quad (10.2)$$

is a bijection. The visible rule is recovered by projection:

$$U = \pi_Y \circ \iota_U. \quad (10.3)$$

Thus every deterministic visible rule is the projection of an invertible map onto its graph.

Proof. The inverse $\text{Graph}(U) \rightarrow X$ is projection onto the second coordinate. The projection identity is immediate. $\square$

For a sequence of rules, the second coordinate may be stored in a persistent history stack or represented by Bennett-style reversible computation [18]. If states are encoded as finite bit strings, standard reversible gates provide a global permutation on a sufficiently large register. The graph theorem isolates the information-theoretic content without committing to one gate set.

10.3 How much ancilla is necessary?

Proposition 10.2 (Worst-case ancilla lower bound). *Let X, Y, A be finite, and let $L : X \rightarrow Y \times A$ be injective with $\pi_Y L = U$. Then*

$$|A| \geq \max_{y \in Y} |U^{-1}(y)|. \quad (10.4)$$

Hence any fixed-length auxiliary record needs at least

$$\log_2 \max_y |U^{-1}(y)| \quad (10.5)$$

bits in the worst case.

Proof. For a fixed y , all $x \in U^{-1}(y)$ have the same visible output. Injectivity of L therefore requires distinct auxiliary values in A for those inputs. $\square$

If X is random and the auxiliary record is losslessly prefix-coded conditional on $U(X)$, its expected length is at least $H(X | U(X))$. Reversibility makes the cost of discarded information explicit rather than eliminating it.

10.4 Bregman projections and dual residuals

The most geometrically meaningful ancilla is often not a full history snapshot but the normal component discarded by projection.

Let Φ be a Legendre function on an open convex domain $\Omega \subseteq \mathbb{R}^d$, and let

$$D_\Phi(z, x) = \Phi(z) - \Phi(x) - \langle \nabla \Phi(x), z - x \rangle \quad (10.6)$$

be its Bregman divergence. Let $C \subseteq \Omega$ be nonempty, closed, and convex, and let

$$y = P_C^\Phi(x) = \arg \min_{z \in C} D_\Phi(z, x). \quad (10.7)$$

Theorem 10.3 (Reversible Bregman projection). *Define the dual residual*

$$q = \nabla\Phi(x) - \nabla\Phi(y). \quad (10.8)$$

Then $q \in N_C(y)$, the normal cone of C at y , and

$$\boxed{x = \nabla\Phi^*(\nabla\Phi(y) + q).} \quad (10.9)$$

Consequently the map

$$x \mapsto (y, q) \quad (10.10)$$

is injective and has the displayed inverse on its image.

Proof. First-order optimality for minimizing $D_\Phi(\cdot, x)$ over C is

$$0 \in \nabla\Phi(y) - \nabla\Phi(x) + N_C(y), \quad (10.11)$$

so $q \in N_C(y)$. Since Φ is Legendre, $\nabla\Phi$ is injective and $\nabla\Phi^*$ is its inverse on the corresponding dual domain. Therefore

$$\nabla\Phi^*(\nabla\Phi(y) + q) = \nabla\Phi^*(\nabla\Phi(x)) = x. \quad (10.12)$$

□

For Euclidean projection, $\Phi(x) = \frac{1}{2} \|x\|^2$ and $q = x - y$. A projection that visibly discards a normal residual is therefore exactly reversible when the residual is propagated.

10.5 Dykstra variables as reversible projection ancillas

Bregman Dykstra algorithms attach dual correction variables to successive constraint projections. These variables carry forward the components that naive cyclic projection would discard. Theorem 10.3 gives a direct interpretation:

$$\boxed{\text{a Dykstra correction is a reusable dual residual of a lossy projection.}} \quad (10.13)$$

In the coupling-dilemma holonomy theorem, the infinite family of carried increments resums the difference between cyclic and joint projection. In reversible PLN, propagating these residuals turns “evidence debt” into explicit state rather than hidden path dependence.

10.6 Exact group-commutator holonomy

Let A and B be invertible lifted rule maps on a microstate space $\tilde{X}$. We use the convention that ABz means apply B and then A .

Definition 10.4 (Rule commutator holonomy). Define

$$H_{A,B} = (BA)^{-1}AB = A^{-1}B^{-1}AB. \quad (10.14)$$

Then

$$AB = BAH_{A,B}. \quad (10.15)$$

Theorem 10.5 (Exact order-defect identity). *If A and B are isometries of $(\tilde{X}, d)$, then for every z ,*

$$\boxed{d(ABz, BAz) = d(H_{A,B}z, z)}. \quad (10.16)$$

More generally, for invertible isometric composite derivations $\tilde{U}_p$ and $\tilde{U}_q$ with common endpoints, let

$$H_{p,q} = \tilde{U}_q^{-1} \tilde{U}_p. \quad (10.17)$$

Then

$$d(\tilde{U}_p z, \tilde{U}_q z) = d(H_{p,q} z, z). \quad (10.18)$$

Proof. Since $AB = BAH_{A,B}$ and BA is an isometry,

$$d(ABz, BAz) = d(BAH_{A,B}z, BAz) = d(H_{A,B}z, z). \quad (10.19)$$

The path statement is identical with BA replaced by $\tilde{U}_q$. $\square$

This identity turns a heuristic “commutator penalty” into the displacement generated by an actual closed executable loop: perform one order, undo the other order, and measure what remains.

10.7 Visible forgetting as projected reversible holonomy

Let $\Pi : \tilde{X} \rightarrow X$ forget residuals, provenance witnesses, inverse frames, and history registers.

Definition 10.6 (Visible path-dependent forgetting). For two lifted derivations p, q , define

$$\mathcal{F}_{\text{vis}}(p, q; z) = \frac{1}{2} d_X(\Pi \tilde{U}_p z, \Pi \tilde{U}_q z)^2. \quad (10.20)$$

Theorem 10.7 (Projection bound). *If Π is nonexpansive and the lifted transports are isometries, then*

$$\boxed{\mathcal{F}_{\text{vis}}(p, q; z) \leq \frac{1}{2} d_{\tilde{X}}(H_{p,q} z, z)^2}. \quad (10.21)$$

Equality holds whenever Π preserves the relevant holonomy displacement isometrically.

Proof. By nonexpansiveness and the exact order-defect identity,

$$d_X(\Pi \tilde{U}_p z, \Pi \tilde{U}_q z) \leq d_{\tilde{X}}(\tilde{U}_p z, \tilde{U}_q z) \quad (10.22)$$

$$= d_{\tilde{X}}(H_{p,q} z, z). \quad (10.23)$$

Square and divide by two. $\square$

Nothing need be destroyed in the total microstate. Visible forgetting can arise because information has moved into hidden residual coordinates or because nontrivial holonomy rotates the full state around a proof loop. Reversibility makes both effects recoverable and auditable, but it does not make the connection flat.

10.8 Reversible idempotent merge

The visible union $E_1 \cup E_2$ is not invertible: it forgets which source supplied which token. A reversible evidence merge retains parent capsules and an overlap witness. At the simplest level,

$$(E_1, E_2) \mapsto (E_1 \cup E_2, E_1, E_2) \quad (10.24)$$

is injective. A content-addressed DAG stores the parents without copying unchanged token payloads. Thus visible confidence can use an idempotent union while the hidden state remains exactly reversible.

11 Reversible HoTT microphysics and directed PLN macrophysics

The reversible dilation suggests a two-level semantic organization.

11.1 The microstate groupoid

Let $\tilde{\mathcal{B}}$ be a type or infinity-groupoid of complete PLN microstates, including visible beliefs, evidence provenance, Bregman/Dykstra residuals, transport frames, and any random seeds needed for replay. A lifted rule instance is an equivalence

$$\tilde{r} : \tilde{x} \simeq \tilde{y}. \quad (11.1)$$

It may therefore be represented by an ordinary HoTT path, with the PGM–HoTT positive/negative and geodesic grades attached.

Let

$$\Pi : \tilde{\mathcal{B}} \longrightarrow \mathcal{B}_{\text{vis}} \quad (11.2)$$

be a forgetting map or modality. The visible rule

$$r : \Pi\tilde{x} \Rightarrow \Pi\tilde{y} \quad (11.3)$$

need not be invertible because its inverse requires the forgotten registers.

Thus

$$\boxed{\text{directed inference} = \text{reversible path} + \text{a forgetting modality}.} \quad (11.4)$$

11.2 Literal proof loops

Two lifted derivations

$$\tilde{p}, \tilde{q} : \tilde{x} \simeq \tilde{y} \quad (11.5)$$

produce the literal loop

$$\tilde{q}^{-1} \cdot \tilde{p} : \tilde{x} \simeq \tilde{x}. \quad (11.6)$$

An evidence fiber over $\tilde{\mathcal{B}}$ therefore carries a monodromy representation

$$\rho_{\mathcal{E}} : \Omega(\tilde{\mathcal{B}}, \tilde{x}) \longrightarrow \text{Aut}(\text{Ev}(\tilde{x})). \quad (11.7)$$

Order-dependent forgetting is the displacement caused by this action.

Remark 11.1 (Reversibility is not flatness). A family of reversible maps may be highly noncommutative. Reversibility makes curvature exactly executable, measurable, and undoable; causal modularity or explicit connection correction is still required to make holonomy small.

11.3 TransWeave as a pseudofunctor on proof space

A TransWeave W should lift from a mapping between SENF representations to a pseudofunctor

$$W_* : \text{Der}(S_1) \longrightarrow \text{Der}(S_2) \quad (11.8)$$

and an evidence map

$$W_{\mathcal{E}} : \text{Ev}(S_1) \longrightarrow \text{Ev}(S_2). \quad (11.9)$$

For every derivation p , require a naturality 2-cell

$$\theta_p : W_{\mathcal{E}} T_p \Rrightarrow T_{W_* p} W_{\mathcal{E}}. \quad (11.10)$$

Its defect

$$\omega_W(p; e) = d_{\mathcal{E}}(W_{\mathcal{E}} T_p e, T_{W_* p} W_{\mathcal{E}} e) \quad (11.11)$$

measures TransWeave coherence curvature. Zero defect means “infer then translate” and “translate then infer” preserve evidence exactly. Alternative weaves that differ syntactically but have zero evidence defect belong to one evidential component and must not be counted as independent support.

12 Atlas-indexed dE–PGM–HoTT for π PLN

A complete π PLN semantics organizes directed higher proof spaces over a context atlas. Each context carries a proof base $\mathcal{B}_C$ and an evidence fibration $\mathcal{E}_C$; the total construction preserves the fiberwise de-duplication and holonomy structure while controlling translation, chart selection, revision, and gluing.

12.1 The context category and guarded translations

Let $\mathcal{C}$ be a category of admissible contexts. A morphism

$$\tau : C \longrightarrow D \quad (12.1)$$

is a guarded translation carrying a declared domain of statements, concept-boundary map, assumption transformation, reliability grade, and untranslated residue. A genuinely partial translation may be represented by a span

$$C \longleftarrow C_0 \longrightarrow D \quad (12.2)$$

from the subcontext C_0 on which it is defined. This avoids pretending that every ontology map is total.

For each context C , let $\mathcal{B}_C$ be the directed higher proof space valid in that context. An admissible translation induces a partial higher functor

$$\tau_* : \mathcal{B}_C \dashrightarrow \mathcal{B}_D. \quad (12.3)$$

We assume these assignments are pseudofunctorial on their domains: identity and composition are respected up to specified higher cells rather than silently identified.

Definition 12.1 (Total atlas proof category). The total context–proof space is the Grothendieck construction

$$\mathcal{B}_{\text{tot}} = \int_{C \in \mathcal{C}} \mathcal{B}_C. \quad (12.4)$$

An object is a pair (C, x) . A 1-cell

$$(\tau, p) : (C, x) \longrightarrow (D, y) \quad (12.5)$$

consists of a guarded translation $\tau : C \rightarrow D$ and a derivation $p : \tau_*x \Rightarrow y$ inside D .

Within-context rule application is the special case $\tau = \text{Id}_C$. A pure context change has p an identity in the target fiber. The total space therefore records whether a cognitive move changed a belief, changed the chart in which the belief is interpreted, or did both.

12.2 Doubly indexed evidence

For each (C, x) let $\text{Ev}_C(x)$ be the provenance-sensitive evidence algebra. The atlas evidence structure is the composite fibration

$$\mathcal{E}_{\text{tot}} = \int_{C \in \mathcal{C}} \mathcal{E}_C \longrightarrow \mathcal{B}_{\text{tot}} \longrightarrow \mathcal{C}. \quad (12.6)$$

A total path $P = (\tau_n, p_n) \cdots (\tau_1, p_1)$ induces evidence transport

$$T_P = T_{p_n} T_{\tau_n} \cdots T_{p_1} T_{\tau_1}, \quad (12.7)$$

with domains understood. Context transport must preserve primitive provenance identity even when it changes statement labels or count valuations.

The persistent π PLN state (4.1) is a section of this structure over statement–context pairs. It need not be a global probability measure; it is a global section of an evidence packet fibration.

12.3 Charts as a contravariant prestack

Inference transport is naturally covariant, but chart restriction is contravariant. We therefore also use a partial prestack

$$\text{Chart} : \mathcal{C}^{\text{op}} \dashrightarrow \text{ProbChart}. \quad (12.8)$$

A refinement or inclusion $i : U \rightarrow C$ restricts a chart Γ_C to $i^*\Gamma_C$ on U . Descent asks whether compatible local charts are restrictions of one chart on a covering context.

This two-direction organization is deliberate:

$\tau_* : \text{push statements and evidence along guarded translations,}$
 $i^* : \text{restrict charts and probabilistic structure to overlaps.}$

(12.9)

12.4 Local Bayesian charts as reversible lenses

Let a_C denote the assumptions and prior parameters of chart C . A chart episode has an import map

$$I_C^{a_C} : \text{Ev}_C \longrightarrow \text{Prob}_C \quad (12.10)$$

and an export map

$$O_C^{a_C} : \text{Prob}_C \longrightarrow \text{Ev}_C. \quad (12.11)$$

They form a well-behaved evidence lens when

$$O_C^{a_C} I_C^{a_C} = \text{Id}_{\text{Ev}_C} \quad (12.12)$$

on the admissible empirical evidence domain. The reverse law need only hold on the image of import, because a chart may contain auxiliary posterior structure not representable by the persistent sufficient statistic.

Theorem 12.2 (Beta chart round trip and prior non-duplication). *Fix $a_C = (\alpha_0, \beta_0)$ and define*

$$I_C^{a_C}(n^+, n^-) = \text{Beta}(\alpha_0 + n^+, \beta_0 + n^-), \quad (12.13)$$

$$O_C^{a_C}(\text{Beta}(\alpha, \beta)) = (\alpha - \alpha_0, \beta - \beta_0) \quad (12.14)$$

on $\alpha \geq \alpha_0, \beta \geq \beta_0$. Then

$$O_C^{a_C} I_C^{a_C}(n^+, n^-) = (n^+, n^-). \quad (12.15)$$

Consequently, any finite number of chart import-export cycles leaves persistent empirical evidence unchanged. By contrast, naively persisting posterior parameters as fresh counts adds one copy of (α_0, β_0) per cycle.

Proof. Substitution gives

$$(\alpha_0 + n^+ - \alpha_0, \beta_0 + n^- - \beta_0) = (n^+, n^-). \quad (12.16)$$

Iteration preserves the same pair. If posterior parameters are instead treated as empirical counts, one cycle maps (n^+, n^-) to $(n^+ + \alpha_0, n^- + \beta_0)$; induction gives an added $m(\alpha_0, \beta_0)$ after m cycles. $\square$

Corollary 12.3 (Charting is not observing). *A pure chart construction and local calculation do not mint primitive evidence tokens. Only an explicit observation or externally justified empirical packet may enlarge provenance support.*

If import or export is lossy, use the reversible graph dilation of Theorem 10.1. There is an injective lift

$$\tilde{I}_C : (e, a_C, 0) \longleftrightarrow (\pi_C, q_C, a_C), \quad (12.17)$$

where q_C is the complement needed to reconstruct e . For exponential-family projection, q_C may be a Bregman dual residual; for a compressed factor graph it may contain discarded sufficient statistics and normalization data.

12.5 Context-guarded evidence equivalence

Parallel proofs from different contexts are not directly comparable. Let

$$p : (C, x) \Rightarrow (C, y), \quad q : (D, x') \Rightarrow (D, y'). \quad (12.18)$$

A comparison requires a covering decision context C_* and guarded translations

$$\tau_C : C \rightarrow C_*, \quad \tau_D : D \rightarrow C_*. \quad (12.19)$$

Definition 12.4 (Cross-context evidence equivalence). Write

$$p \simeq_{\mathcal{E}^*} q \quad (12.20)$$

when the translated endpoints are identified in C_* and a zero-defect 2-cell witnesses

$$T_{\tau_C} T_p = T_{\tau_D} T_q \quad (12.21)$$

on the compared provenance support.

Theorem 12.5 (Guarded cross-context de-duplication). *If $p \simeq_{\mathcal{E}^*}^C q$, context transports preserve joins, and the evidence algebra in C_* is idempotent, then*

$$T_{\tau_C} T_p(e) \vee T_{\tau_D} T_q(e) = T_{\tau_C} T_p(e). \quad (12.22)$$

If no admissible covering context and translation witness exist, the cross-context merge is undefined rather than automatically additive.

Proof. The evidence-equivalence witness makes the two transported elements equal. Idempotence gives the displayed identity. The final clause is a typing rule: elements of distinct context fibers have no primitive merge operation until transported to a common fiber. $\square$

This rule blocks two symmetric errors. It prevents ontology smear, in which equal surface strings with different concept boundaries are averaged. It also prevents false independence, in which two chart-specific presentations of one primitive observation are counted twice.

12.6 Homotopy descent for probabilistic charts

Let $\{U_i \rightarrow U\}$ be a covering family of contexts and Γ_i local charts. On overlaps U_{ij} suppose there are graded positive identities

$$\alpha_{ij} : \Gamma_i|_{U_{ij}} \simeq \Gamma_j|_{U_{ij}} \quad (12.23)$$

with bounded negative-identity evidence and higher coherence on triple overlaps.

Definition 12.6 (Chart descent type). The type of coherent global gluings is

$$\begin{aligned} \text{Desc}_U(\Gamma_\bullet) := & \sum_{\Gamma_U} \prod_i \text{Id}^+(\Gamma_U|_{U_i}, \Gamma_i) \\ & \times \prod_{i,j,k} \text{Coh}(\alpha_{jk} \circ \alpha_{ij}, \alpha_{ik}) \times \text{Cons}^-. \end{aligned} \quad (12.24)$$

The PGM–HoTT definiteness criterion now applies one level higher.

Proposition 12.7 (Geodesic gluing trichotomy). *For a chart family $\Gamma_\bullet$:*

- (i) *if $\text{Desc}_U(\Gamma_\bullet)$ is geodesically contractible, there is an essentially unique acceptable global chart;*
- (ii) *if it is inhabited but not geodesically contractible, several inequivalent global charts remain and must be guarded as alternatives;*
- (iii) *if it is empty, the local charts are non-gluable in the chosen chart class.*

Forgetting grades recovers the ordinary homotopy-descent statements.

Proof. Item (i) is the PGM–HoTT soundness of geodesic definiteness applied to the descent type. Item (ii) is merely existence without uniqueness. Item (iii) is the definition of failure of descent. The grade-forgetting modality preserves the relevant finite limits and identity data by the PGM–HoTT assumptions. $\square$

For the anti-correlation family (4.17), the descent type of global joint probability charts is empty even though all one-variable overlap data agree.

12.7 A quantitative descent obstruction

Let $r_i\Gamma$ denote restriction of a candidate global chart to U_i , and let d_i be separating chart metrics or divergences. Define

$$\mathcal{O}_{\text{desc}}(\Gamma_\bullet) = \inf_{\Gamma_U} \left(\sum_i \omega_i d_i(r_i\Gamma_U, \Gamma_i)^2 \right)^{1/2}. \quad (12.25)$$

Proposition 12.8 (Zero obstruction and exact gluing). *If an exact global chart exists, then $\mathcal{O}_{\text{desc}} = 0$. Conversely, if the infimum in (12.25) is attained and every d_i separates equality, then $\mathcal{O}_{\text{desc}} = 0$ implies an exact gluing.*

Proof. An exact global chart makes every summand zero. Conversely, an attaining chart with zero weighted sum has $d_i(r_i\Gamma_U, \Gamma_i) = 0$ for every positive weight, hence exact equality on every chart; the descent coherence data then supply the gluing. $\square$

The obstruction depends on the proposed global chart class. Enlarging the model class can reduce it; this is a feature, because ontology or layer revision may create a richer covering context. It must nevertheless be reported rather than hidden inside a forced approximate posterior.

12.8 Three kinds of local coherence cell

The atlas introduces new two-dimensional defects in addition to ordinary proof squares.

For a rule r in C and a translation $\tau : C \rightarrow D$, define the inference–translation naturality defect

$$\delta_{\tau,r}(e) = d_{\mathcal{E}_D}(T_\tau T_r(e), T_{\tau_* r} T_\tau(e)). \quad (12.26)$$

It compares “infer then translate” with “translate then infer.”

For $C \xrightarrow{\tau} D \xrightarrow{\sigma} E$, define the context-composition defect

$$\delta_{\sigma,\tau}(e) = d_{\mathcal{E}_E}(T_{\sigma \circ \tau}(e), T_\sigma T_\tau(e)). \quad (12.27)$$

For a chart lens and a revision R , define

$$\epsilon_C(e) = d_{\mathcal{E}_C}(O_C I_C e, e), \quad \delta_R(e) = d(T_{R_{\text{specified}}} e, T_{R_{\text{implemented}}} e). \quad (12.28)$$

The beta chart theorem gives $\epsilon_C = 0$ in its exact domain.

12.9 Atlas local-to-global coherence

Theorem 12.9 (Atlas proof-surface bound). *Let $P, Q : (C, x) \rightarrow (D, y)$ be total paths in $\mathcal{B}_{\text{tot}}$ connected by a finite higher surface Σ . Suppose its generating cells are proof-rewrite cells, inference–translation cells, context-composition cells, chart round-trip cells, and revision cells. Assume all common pre- and post-compositions are nonexpansive. Then*

$$\begin{aligned} d_{\mathcal{E}_D}(T_P e, T_Q e) &\leq \sum_{\sigma \in \Sigma_{\text{proof}}} \delta_\sigma + \sum_{(\tau,r) \in \Sigma_{\text{mixed}}} \delta_{\tau,r} \\ &+ \sum_{(\sigma,\tau) \in \Sigma_{\text{ctx}}} \delta_{\sigma,\tau} + \sum_{C \in \Sigma_{\text{chart}}} \epsilon_C + \sum_{R \in \Sigma_{\text{rev}}} \delta_R. \end{aligned} \quad (12.29)$$

Proof. Choose a sequence of total paths from P to Q replacing one generating cell at a time. The change caused by each replacement is bounded by the cell defect after common prefixes and suffixes, by nonexpansiveness. Sum with the triangle inequality. This is the same adjacent-replacement proof as the single-context surface theorem, now in the Grothendieck construction. $\square$

Corollary 12.10 (Flat atlas). *If every generating defect in (12.29) vanishes, chart lenses obey exact round-trip laws, and the relevant descent type is inhabited, then evidence transport is independent of proof and chart presentation on every contractible total surface. If the descent type is also geodesically contractible, the global chart is essentially unique.*

12.10 Smooth total curvature and exact total holonomy

Assume the total atlas bundle admits a smooth horizontal/vertical splitting. The curvature two-form decomposes by bidegree:

$$R^{\text{tot}} = R_{\text{proof}}^{2,0} + R_{\text{mixed}}^{1,1} + R_{\text{context}}^{0,2}. \quad (12.30)$$

The first component measures rule-order curvature inside a context. The second measures failure of inference to be natural under context translation. The third measures curvature of chart and ontology transport.

The decomposition is a splitting of the curvature form, not an assertion that finite effects add as commuting scalars. The exact finite transport is the nonabelian surface-ordered exponential of the full form.

Theorem 12.11 (Total relative holonomy). *Let $\tilde{T}_P$ and $\tilde{T}_Q$ be reversible, metric-compatible lifts of two total paths with common endpoints. Define*

$$H_{P,Q} = \tilde{T}_Q^{-1} \tilde{T}_P. \quad (12.31)$$

Then

$$\boxed{\mathcal{F}_{\text{atlas}}(P, Q; e) = \frac{1}{2} \|(H_{P,Q} - I)e\|^2.} \quad (12.32)$$

If the resulting loop bounds a surface Σ , then

$$H_{P,Q} = \text{Sexp} \int_{\Sigma} R^{\text{tot}}. \quad (12.33)$$

Proof. Metric compatibility gives

$$\|\tilde{T}_P e - \tilde{T}_Q e\| = \|\tilde{T}_Q(H_{P,Q} e - e)\| = \|(H_{P,Q} - I)e\|. \quad (12.34)$$

The surface formula is nonabelian Stokes for the total connection. $\square$

The rank-two and multi-angle $\sin^2$ laws from the single-context theory apply to $H_{P,Q}$ on the reversible residual subbundle.

12.11 Holonomy and descent obstruction are independent

Proposition 12.12 (Two distinct global defects). *Neither of the following implications holds in general:*

$$R^{\text{tot}} = 0 \implies \text{Desc}_U(\Gamma_{\bullet}) \text{ inhabited}, \quad (12.35)$$

$$\text{Desc}_U(\Gamma_{\bullet}) \text{ inhabited} \implies H_{P,Q} = I. \quad (12.36)$$

Proof. For the first failure, take the three anti-correlated pairwise charts. Within each chart and on each one-variable overlap, choose trivial flat transport. The curvature vanishes locally, but the global-joint descent type is empty by the non-extendibility theorem. For the second failure, take one globally defined probability chart but two nonlinear, noncommuting evidence projections inside it. The chart glues trivially, while their reversible group commutator is nonidentity. $\square$

Thus a complete diagnostic must report both path transport and chart existence.

12.12 Multiplicity, holonomy, and descent

Definition 12.13 (Atlas inference defect). For a task-specific family of local charts and two complete context-proof paths, define

$$\mathcal{E}_{\pi\text{PLN}} = \mathcal{E}_{\text{mult}} + \frac{1}{2} \|(H_{P,Q} - I)e\|^2 + \lambda_{\text{desc}} \mathcal{O}_{\text{desc}}(\Gamma_{\bullet})^2 + E_{\text{approx}}. \quad (12.37)$$

The first term measures wrong effective use of unique evidence. The second measures path-dependent proof/context transport. The third measures failure of the selected probability charts to admit a coherent global realization in the proposed model class. The expression is a diagnostic and control objective in general. It becomes a literal error decomposition only under additional orthogonality and smooth reduction assumptions that identify the three residual channels.

12.13 Revision as a first-class higher morphism

A visible revision has the form

$$R : (C, H, E) \longrightarrow (C', K). \quad (12.38)$$

It may add evidence, discount provenance, quarantine a packet, change a chart, split a concept, move to another ontology, or alter the valuation that defines weakness and relevance. Its grade can record

$$\ell(R) = (\text{fit}, \text{pres}, \text{weak}, \text{cost}, \text{prov}, \text{coh}). \quad (12.39)$$

The preferred revision is selected by (4.22)–(4.23).

Proposition 12.14 (Reversible revision lift). *Every deterministic revision $R : S \rightarrow S'$ has an injective graph lift*

$$\tilde{R}(s, 0) = (R(s), s). \quad (12.40)$$

For packet-level π PLN revision, the retained complement may be restricted to the information actually overwritten: old reliability and activity flags, old context and statement labels, the previous translation witness, the old chart assumptions, and any discarded packet decomposition.

Proof. The graph lift is injective because its second component recovers s . The listed smaller complement is sufficient whenever the visible revision and unchanged persistent DAG together reconstruct all other fields. $\square$

A revision loop can therefore possess holonomy even when every step is auditable and undoable. Reversibility makes the defect recoverable; it does not make revision order-independent.

13 Hybrid geodesic control on the reversible π PLN atlas

13.1 Local bridge actions, atlas moves, and fiber constraints

Within a fixed context C , the Schrodinger-bridge action lives on a local probabilistic base and has a local factorization

$$\rho_C(x, t) \propto f_C(x, t)g_C(x, t). \quad (13.1)$$

Across contexts there need not be one probability density. A complete atlas path alternates local bridge segments with guarded translations and revision moves:

$$\gamma = \gamma_1 \xrightarrow{\tau_1} \gamma_2 \xrightarrow{R_2} \gamma_3 \xrightarrow{\tau_3} \dots \quad (13.2)$$

Its semantic label is naturally product-quantale-valued:

$$\mathcal{L}(\gamma) = \bigotimes_i \mathcal{L}_{\text{SB}}^{C_i}(\gamma_i) \otimes \bigotimes_j \ell(\tau_j) \otimes \bigotimes_k \ell(R_k). \quad (13.3)$$

A scalar action is a scheduling projection of this richer label, not the foundational semantics.

Evidence consistency lives in the total fiber. The local bridge energy, provenance constraints, revision quality, and descent risk need not be identified.

Let $e(t)$ be an evidence section along a controlled path and $D_t^\mathcal{E} e$ its covariant derivative. Let J_{obs} denote genuinely new observational evidence. A useful scalarized augmented objective is

$$\begin{aligned} \mathcal{A}_{\text{tot}} = & \mathcal{A}_{\text{SB}}[\rho, v] + \frac{\eta}{2} \int \rho \left\| D_t^\mathcal{E} e - J_{\text{obs}} \right\|^2 dt \\ & + \lambda_{\text{dc}} \int \rho \mathcal{D}_{\text{dc}} dt + \lambda_{\text{curv}} \sum_{\sigma \in \Sigma} \delta_\sigma + \lambda_{\text{desc}} \mathcal{O}_{\text{desc}}(\Gamma_\bullet)^2. \end{aligned} \quad (13.4)$$

The terms respectively choose low-effort local proof corridors, enforce lawful evidence transport, penalize repeated provenance, discourage locally curved rule/context combinations, and penalize forced use of a non-gluable chart family.

An alternative is to minimize only $\mathcal{A}_{\text{SB}}$ subject to hard constraints

$$D_t^\mathcal{E} e = J_{\text{obs}}, \quad \mathcal{D}_{\text{dc}} = 0, \quad \mathcal{F}_{\text{order}} \leq \varepsilon_{\text{hol}}. \quad (13.5)$$

This makes clear that path effort, evidence conservation, and holonomy are distinct structures.

Proposition 13.1 (Path-independent evidence in the flat exact limit). *Suppose a simply connected region of proof space has zero evidence curvature on every generating 2-cell, transports preserve idempotent joins, and pure inference introduces no new tokens. Then any two homotopic derivations with common endpoints transport the same evidence, and merging all such derivations adds no confidence beyond one representative.*

Proof. Zero cell defects and the proof-surface theorem imply $T_p = T_q$ for homotopic paths. Exact homotopy de-duplication then gives an idempotent merge. $\square$

13.2 Curvature-, revision-, and evidence-aware BGC

For a partial atlas path P in context C and a candidate action a , where a may be a rule, chart transition, or revision, define:

- $\Delta I_{\text{uniq}}(a \mid P)$: unique provenance newly delivered by a ;
- $I_{\text{reuse}}(a \mid P)$: provenance already represented on P and reused by a ;
- $\delta_{\text{coh}}(a, P)$: proof, mixed, context, chart, and revision cell defects created by a ;
- $\text{fit}(a)$ and $\text{pres}(a)$: residuated revision scores when a changes a state or context;
- $\text{weak}(a)$: task-relative weakness or generality;
- $\Delta \mathcal{O}_{\text{desc}}(a)$: predicted change in chart-gluing obstruction.

Inside a local chart, a practical scalar priority is

$$\begin{aligned} \text{Score}(a \mid P, C) = & \Delta(\log f_C + \log g_C) - \lambda C_a \\ & + \beta \Delta I_{\text{uniq}}(a \mid P) - \eta I_{\text{reuse}}(a \mid P) \\ & - \zeta \delta_{\text{coh}}(a, P) + \rho \text{fit}(a) + \sigma \text{pres}(a) \\ & + \nu \text{weak}(a) - \chi \Delta \mathcal{O}_{\text{desc}}(a). \end{aligned} \quad (13.6)$$

For a pure within-chart rule, the revision and context terms may be neutral. For a chart-selection or ontology-revision move, $f_C g_C$ is replaced by the appropriate boundary-value or learned task-value estimate on the candidate context.

This score does not quotient away useful path multiplicity. Multiple proof or chart routes may improve robustness and reachability. It prevents the controller from treating their multiplicity as independent evidence, from crossing highly nonnatural translations without paying for them, or from forcing a globally incoherent chart merely because its local posterior is sharp.

13.3 Covering-context and holonomy-aware splicing

When forward and backward strands meet in the same context, syntactic unification is insufficient; their transported evidence states should agree. When they meet in different contexts C_F and C_B , even that comparison is ill typed until the controller constructs a covering context C_* with translations

$$\tau_F : C_F \rightarrow C_*, \quad \tau_B : C_B \rightarrow C_*. \quad (13.7)$$

Let

$$e_F^* = T_{\tau_F} e_F, \quad e_B^* = T_{\tau_B} e_B. \quad (13.8)$$

Define

$$\mathcal{H}_{\text{splice}} = \frac{1}{2} d_{\mathcal{E}_{C_*}}(e_F^*, e_B^*)^2 + \lambda_{\text{desc}} \mathcal{O}_{\text{desc}}(\Gamma_F, \Gamma_B; C_*)^2. \quad (13.9)$$

A splice score may be

$$\text{SpliceScore} = \log f_{C_F} + \log g_{C_B} - \lambda C - \xi \mathcal{H}_{\text{splice}}. \quad (13.10)$$

If no admissible covering context exists, the strands do not splice merely because their surface strings unify.

In the reversible lift, compute the exact relative loop after transporting both strands to C_* :

$$H_{F,B} = \tilde{T}_{P_B^*}^{-1} \tilde{T}_{P_F^*}, \quad (13.11)$$

and penalize its displacement on the relevant evidence subspace.

13.4 Dykstra-PLN

A Dykstra-corrected PLN rule application transports:

$$\text{visible truth value} + \text{provenance capsule} + \text{dual projection residual}. \quad (13.12)$$

The residual is passed to subsequent projections rather than discarded. At atlas level, analogous residuals accompany chart import/export, context translation, and revision. A proof-and-chart cycle then carries an explicit evidence and context debt whose fixed point reconstructs the joint overlap-corrected, prior-unwound projection when such a projection exists. This is the natural reversible counterpart of cyclic π PLN inference.

14 MORK realization

The MORK content-addressed DAG proposed in geodesic inference control already supplies immutable objects, parent CIDs, deterministic merge frontiers, and local evidence sketches [1]. The following records add higher proof cells and reversible state.

14.1 Rewrite cells and proof classes

```
type: RewriteCell
fields: {
  lhs_path_cids[],
  rhs_path_cids[],
  source_state_cid,
  target_state_cid,
  rewrite_kind,
  semantic_identity_cid,
  lhs_capsule_cid,
  rhs_capsule_cid,
  evidence_distance,
  overlap_delta,
  coherence_grade,
  parents[]
}
```

```
type: EvidenceProofClass
fields: {
  endpoint_atom_id,
  representative_path_cid,
  member_path_digest,
  evidence_capsule_cid,
  evidence_sketch_cid,
  positive_weight,
  negative_weight,
  best_control_cost,
  aggregate_fg_mass,
  rewrite_cell_cids[]
}
```

```
}
```

The fields ‘aggregate_fg_mass’ and evidence weight are intentionally separate. Many search paths may increase the first while idempotent provenance leaves the second unchanged.

14.2 Reversible rule applications

```
type: ReversibleRuleApp
fields: {
    rule_id,
    binding_uid,
    premise_capsule_cids[],
    prior_output_capsule_cid,
    output_capsule_cid,
    exact_overlap_witness_cid,
    bregman_residual_cid,
    dykstra_increment_cid,
    transport_frame_cid,
    rng_seed,
    parent_ruleapp_cid,
    inverse_checksum
}
```

Persistent sharing means that unchanged state need not be copied. Storage is concentrated on information actually lost by the visible rule: overwritten pointers, overlap decomposition, projection residuals, and nondeterministic choices.

14.3 Context, chart, descent, and revision records

The atlas layer adds explicit context and chart objects rather than hiding them in rule metadata.

```
type: ChartLensRecord
fields: {
    context_cid,
    chart_cid,
    assumption_package_cid,
    local_prior_cid,
    import_map_id,
    export_map_id,
    chart_complement_cid,
    roundtrip_error,
    parent_cids[]
}
```

```
type: ContextTransition
fields: {
    source_context_cid,
    target_context_cid,
```

```

        translation_map_cid,
        reliability_grade,
        untranslated_residue_cid,
        inverse_complement_cid,
        naturality_probe_cids[],
        parent_cids[]
    }

type: DescentCell
fields: {
    local_chart_cids[],
    overlap_context_cids[],
    transition_cids[],
    pairwise_compatibility_grades[],
    higher_coherence_grade,
    global_chart_candidate_cid,
    gluing_defect,
    parent_cids[]
}

type: RevisionEdge
fields: {
    old_state_cid,
    evidence_packet_cid,
    revised_state_cid,
    source_context_cid,
    target_context_cid,
    fit_grade,
    preservation_grade,
    weakness_grade,
    cost_grade,
    correction_kind,
    reversible_complement_cid,
    parent_cids[]
}

```

A practical inferred Atom or statement record should therefore carry at least

```

Statement, (n_plus, n_minus), STVProjection,
ContextID, ChartID, AssumptionPackageCID, LocalPriorCID,
ProvenanceCID, Weakness, TaskTrace, Attention,
TransportFrameCID, DescentStatus

```

with the scalar STV treated as a cached interface rather than the primary evidence object.

14.4 Local curvature probes

For a candidate pair r, s , execute both orders in a temporary namespace:

```
capsule_rs = apply(s, apply(r, capsule))
capsule_sr = apply(r, apply(s, capsule))
```

```
delta_rs = evidence_distance(capsule_rs, capsule_sr)
```

Cache the result by rule schemas, binding-overlap class, context class, and evidence-overlap bucket. Only neighbors in a sparse confusion graph require probes. Under the assumptions of the proof-surface theorem, the extra per-step work is proportional to the local confusion degree rather than total proof length.

14.5 Atlas frontier expansion

```
expand(path P, action a):
```

```
    C_target = choose_target_context(P.context, a)
    transition = guarded_translation(P.context, C_target, a)

    if transition is undefined:
        reject(a)

    chart = retrieve_or_build_chart(C_target, P.task)
    capsule_new = apply_action_in_chart(a, transition, chart, P.capsule)
    capsule_new = export_prior_unwound(capsule_new, chart)

    unique_gain =
        sketch_difference_weight(capsule_new, P.capsule)

    reused_mass =
        sketch_intersection_weight(capsule_new, P.capsule)

    coherence =
        proof_commutator_cost(a, P)
        + translation_naturality_cost(transition, a, P)
        + chart_roundtrip_cost(chart)
        + revision_coherence_cost(a)

    descent_delta =
        estimate_descent_change(P.chart_family, chart, transition)

    revision_grades =
        fit_preservation_weakness(a, P, C_target)

    proof_class =
        canonicalize_in_covering_context(
            endpoint=capsule_new.atom,
            context=C_target,
            evidence_digest=capsule_new.digest)
```



```

score =
    delta_local_log_f_plus_log_g
    - lambda_cost * step_cost(a)
    + beta_unique * unique_gain
    - eta_reuse * reused_mass
    - zeta_coh * coherence
    + rho_fit * revision_grades.fit
    + sigma_pres * revision_grades.preservation
    + upsilon_weak * revision_grades.weakness
    - chi_desc * descent_delta

enqueue(path_node,
        context=C_target,
        chart=chart,
        control_score=score,
        evidence_class=proof_class)

```

With exact token CIDs, de-duplication is exact. With KMV, HLL, or related sketches, the approximate de-duplication theorem turns sketch error into an explicit confidence-error bound, provided the chosen evidence metric and valuation satisfy the stated Lipschitz assumptions. Chart round-trip error and translation naturality error enter separately, so implementation diagnostics can distinguish prior leakage, ontology mismatch, and genuine rule curvature.

15 Worked examples

15.1 Repeated chart construction does not create evidence

Suppose the persistent store contains empirical counts

$$e = (8, 2) \tag{15.1}$$

and a local chart uses prior $\text{Beta}(1, 1)$. Import gives $\text{Beta}(9, 3)$. Correct export subtracts the prior and returns $(8, 2)$ exactly. Repeating the chart episode any number of times leaves the persistent evidence unchanged.

A naive system that stores posterior parameters as if they were fresh empirical counts produces

$$(8, 2) \mapsto (9, 3) \mapsto (10, 4) \mapsto \dots \tag{15.2}$$

After m cycles it has fabricated m positive and m negative virtual observations. This is evidence double counting caused by confusing a chart coordinate choice with an observation. The reversible lens law (12.15) removes it exactly.

15.2 Concept boundaries and covering contexts

Let context C_A define

$$\text{Tall}_A(x) \equiv \text{Height}(x) > 200 \text{ cm}, \tag{15.3}$$

while C_B defines

$$\text{Tall}_B(x) \equiv \text{Height}(x) > 180 \text{ cm}. \tag{15.4}$$

For a person of height 185 cm, evidence is negative for Tall_A and positive for Tall_B . The two packets are not contradictory measurements and must not be averaged under the bare string “Tall.”

A decision task may construct a covering context C_* containing height as a common measurable quantity and translations

$$\tau_A(\text{Tall}_A) = \text{Height} > 200, \quad \tau_B(\text{Tall}_B) = \text{Height} > 180. \quad (15.5)$$

The packets can now be compared without merging their proposition labels. The covering context explains the apparent conflict as a boundary difference. Without these translations, guarded typing blocks the merge and thereby prevents ontology smear.

15.3 A non-gluable chart triangle

Take the three pairwise charts of (4.17). Each is internally coherent, every one-variable restriction is uniform, and all context-composition transports on those single-variable overlaps can be chosen flat. Nevertheless, no global joint chart exists. Thus

$$R^{\text{context}} = 0 \quad \text{on the represented overlaps,} \quad \mathcal{O}_{\text{desc}} > 0 \quad (15.6)$$

in the class of ordinary joint distributions. A controller asked for one joint answer has several legitimate revision moves: weaken one pairwise chart, enlarge the ontology with a contextuality marker, retain all charts and report non-gluability, or reject the global query as ill posed. Forcing a posterior is not the only coherent response.

15.4 Mixed curvature from infer–translate order

Let a source context classify “fever” by a loose commonsense threshold and a target medical context classify it by a calibrated temperature threshold. A deduction rule r may transform symptom reports into a fever judgment in the source context. Compare

$$T_\tau T_r(e) \quad \text{with} \quad T_{\tau_* r} T_\tau(e). \quad (15.7)$$

If the source rule silently used the loose predicate boundary while the transported rule uses the strict boundary, the two evidence packets differ. The defect $\delta_{\tau, r}$ is neither ordinary within-context rule curvature nor chart non-gluability. It is mixed curvature: inference is not natural under the proposed ontology translation. The appropriate intervention is to refine the translation or covering context, not merely to precondition the truth-value metric.

15.5 Two proof paths, one causal innovation

Let one primitive observation token z support both A and B :

$$z \Rightarrow A, \quad z \Rightarrow B, \quad (15.8)$$

and let two derivations infer C :

$$p : z \Rightarrow A \Rightarrow C, \quad q : z \Rightarrow B \Rightarrow C. \quad (15.9)$$

Suppose a zero-defect rewrite cell witnesses that both routes transport the same token z to C . Then

$$T_p \{z\} = T_q \{z\} = \{z\} \quad (15.10)$$

and

$$T_p \{z\} \vee T_q \{z\} = \{z\}. \quad (15.11)$$

If w_z is the token weight, naive addition gives $2w_z$, whereas idempotent fusion gives w_z . The double-counting functional is

$$\mathcal{D}_{\text{dc}}(\{p, q\}; \{z\}) = w_z. \quad (15.12)$$

Now add an independent observation u and a third derivation r carrying u . Correct fusion is

$$T_p e \vee T_q e \vee T_r e = \{z, u\}, \quad (15.13)$$

with weight $w_z + w_u$. The controller may retain all three paths for search diversity, but the evidence layer has two independent innovations, not three proofs.

15.6 Noncommuting projection rules

Consider Euclidean evidence state $x \in \mathbb{R}^2$. Let P project onto the x -axis and Q project onto a line at angle θ :

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{pmatrix}. \quad (15.14)$$

Then

$$QP - PQ = \frac{\sin 2\theta}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (15.15)$$

The order defect vanishes only when the constraint lines are parallel or orthogonal. Each visible projection is noninvertible, but the reversible Bregman lift carries

$$q_P = x - Px, \quad q_Q = x - Qx. \quad (15.16)$$

The pair (Px, q_P) reconstructs x , as does (Qx, q_Q) . Thus the order-dependent visible endpoint is not caused by fundamental information loss; it is caused by noncommuting constraint geometry. Dykstra corrections propagate the normal residuals needed to recover the joint projection.

15.7 A reversible rule-loop probe

Let A and B be reversible metric-compatible microstate implementations of two PLN operations, for example a provenance-aware revision and a context transport. The exact loop

$$H_{A,B} = A^{-1}B^{-1}AB \quad (15.17)$$

can be run directly. If $H_{A,B}z = z$ on the relevant evidence subspace, the pair is locally flat even if the visible rules look different. If it moves z , the displacement is the exact order defect by (10.16). This gives an empirical curvature probe requiring no global differential-geometric model.

16 Empirical predictions and evaluation program

16.1 Prior-cycling and chart-roundtrip tests

Repeatedly enter and leave the same local beta or factor-graph chart without adding observations. Correct prior-unwound export predicts invariant persistent evidence. A system that persists posterior coordinates will show linear evidence inflation with the number of chart cycles. This isolates chart-induced double counting from proof-path duplication.

16.2 Holonomy versus non-gluability

Construct two benchmark families. In the first, use one global chart with noncommuting nonlinear rule transports: descent obstruction is zero but holonomy is nonzero. In the second, use the anti-correlated pairwise charts with flat overlap transport: local holonomy is zero while descent obstruction is positive. A diagnostic that reports only one scalar should fail to distinguish them; the three-channel diagnostic (12.37) should not.

16.3 Translation naturality predicts ontology errors

For pairs of contexts with known boundary shifts, estimate $\delta_{\tau,r}$ by executing infer-then-translate and translate-then-infer. The prediction is that large mixed defects identify unsafe context translations and correlate with ontology-smear errors better than within-context commutators alone.

16.4 Reversible corrective revision

Apply quarantine, reliability discount, retraction, and relabeling to provenance packets. Verify exact replay from the reversible complement. Compare this with adding counterevidence: the visible aggregate may change similarly, but only the reversible packet-level representation preserves the distinction between “new evidence against the claim” and “old support was invalidly counted.”

The synthesis suggests experiments that separate multiplicity, curvature, and hidden residual effects.

16.5 Path multiplicity versus unique evidence

Construct benchmark proof graphs with many paraphrastic or permuted derivations of the same primitive evidence and a smaller number of genuinely independent observations. Compare:

- (a) raw path-count confidence;
- (b) sketch-corrected confidence;
- (c) exact token-union confidence;
- (d) control mass allocated by the fg scheduler.

The prediction is that raw path count inflates confidence, while exact evidence mass remains invariant under addition of evidence-equivalent paths. Control mass may still increase, validating the need for two measures.

16.6 Local commutators predict global path dependence

Estimate δ_{rs} for sampled rule pairs and build a confusion graph. For alternative proof orders, compare the measured endpoint evidence distance with

$$\sum_{\{r,s\} \in E_{\text{conf}}} \delta_{rs}. \quad (16.1)$$

The prediction is that global order sensitivity tracks the sum over confusing pairs and remains small in sparse causal factorizations even when total proof length is large.

16.7 Exact provenance removes inflation but not curvature

Replace lossy evidence sketches with exact token CIDs. Double-counting error should drop sharply. Residual order dependence should remain when nonlinear rule transports do not commute. This experiment distinguishes provenance error from connection curvature.

16.8 Reversible residuals make forgetting recoverable

For projection-like PLN updates, store dual residuals and compare three operations:

- (i) visible sequential projection;
- (ii) inverse replay using residuals;
- (iii) Dykstra-corrected joint projection.

The reversible state should recover the pre-step evidence exactly up to numerical precision. Dykstra correction should reduce sequential-versus-joint discrepancy without requiring a semantic inverse implication.

16.9 Curvature-aware geodesic scheduling

Compare the baseline score (2.6) with (13.6). Measure proof cost, endpoint confidence, duplicate evidence, splice disagreement, and robustness to rule reordering. The expected benefit is not necessarily a shorter proof. It is a proof whose evidence is more unique, more order-stable, and easier to audit.

16.10 Storage-efficiency frontier

The ancilla bound (10.4) predicts a tradeoff between reversible auditability and storage. For families of increasingly many-to-one truth functions, measure retained bits per step and compare with empirical conditional entropy $H(X \mid U(X))$. Compression should approach but not beat the information-theoretic lower bound without sacrificing exact reversibility.

17 Limitations and open problems

17.1 Choosing the context site and covers

The atlas formalism presupposes a category or restriction structure of contexts and a notion of covering family. In practice these are learned and task-relative. Bad cover choices can hide a valid gluing or manufacture an apparent one. The theory makes the choice explicit but does not solve context discovery by itself.

17.2 Model-class dependence of descent obstruction

A positive $\mathcal{O}_{\text{desc}}$ is relative to a proposed class of global charts and divergences. Enlarging the model class, adding latent variables, or refining the ontology may turn a non-gluable family into a gluable one. The obstruction is therefore a diagnostic of the current representational layer, not an absolute metaphysical claim.

17.3 Partial translations and untranslated residue

Real cross-context maps are partial and may leave semantically important residue. Treating a partial translation as total can make naturality defects look artificially small because untranslatable content has been dropped. Implementations should retain an explicit residue object and charge its loss in the coherence grade.

17.4 Priors, hyperpriors, and chart identity

The beta round-trip theorem assumes the same recorded prior is subtracted on export. If chart priors are revised, learned, or hierarchically coupled, the chart identity and reversible complement must include the prior version and its provenance. Otherwise changing a prior can be mistaken for empirical evidence.

17.5 Formal type theory

The paper supplies a semantic core, not a fully mechanized directed dependent type theory. A complete treatment should define substitution, universes, directed dependent elimination, higher inductive types, interaction with PGM–HoTT modalities, and a normalization or canonicity theorem where appropriate.

17.6 Causal identity of evidence tokens

Exact token identity is not always the same as causal independence. Two sensor records may have different CIDs but be deterministic descendants of one latent event. The ideal evidence algebra should therefore quotient primitive records by a justified causal-equivalence relation or represent a richer causal provenance hypergraph. Learning this relation without erasing genuine independence is a central open problem.

17.7 Smooth reduction validity

The curvature decomposition (9.1) is conditional. Many symbolic PLN rules are piecewise, discrete, or nonprojective. The discrete proof-surface theorem remains valid, but Fisher–Rao curvature may not be the correct smooth invariant. One needs empirical and theoretical criteria for when a differentiable probabilistic-circuit or energy-based realization faithfully approximates the symbolic engine.

17.8 Approximate sketches

Idempotence at the object/CID level does not imply unbiased evidence cardinality under all sketches. False overlap and missed overlap can respectively undercount and overcount evidence. The metric and Lipschitz constants in the approximate de-duplication theorem should be worked out for each sketch family and correlation regime.

17.9 Nonabelian proof holonomy

In rank greater than two, no single curvature scalar determines forgetting for all evidence directions. The complete invariant is the holonomy conjugacy class or its angle spectrum. Efficiently estimating

the leading angles from local rule probes without constructing a huge transport matrix is an open algorithmic problem.

17.10 Privacy and reversibility

Exact reversible logs can retain sensitive information that an ordinary lossy inference would discard. Privacy-preserving reversible computation, encrypted provenance witnesses, retention policies, and deliberate irreversible erasure require explicit design. Landauer’s principle reminds us that erasure is a physical and governance choice, not merely a software omission [19].

18 Conclusion

Geodesic inference control, PGM-HoTT, π PLN, causal coding, and the curvature theory of forgetting fit together once five different structures are given different mathematical homes.

Search probability lives on raw context–proof paths. Semantic and evidential identities live on higher cells. Persistent positive and negative evidence lives in an idempotent, provenance-rich section indexed by context. Ordinary probability lives inside temporary local charts. Revision and context translation are explicit moves rather than invisible changes of interpretation.

This organization exposes three failure modes that a single posterior or scalar confidence cannot distinguish:

$$\begin{array}{l} \text{multiplicity error} = \text{wrong effective use of unique evidence,} \\ \text{holonomy} = \text{path-dependent transport around proof/context loops,} \\ \text{descent obstruction} = \text{failure of local probability charts to glue globally.} \end{array} \quad (18.1)$$

Metric changes and scheduler reweighting address the first. Causal modularity, coherent translations, and connection flattening address the second. Context refinement, ontology revision, latent-variable enrichment, or an explicit report of non-gluability addresses the third. Treating all three as “uncertainty” applies the wrong intervention.

The reversible microstate formulation makes the geometry executable. Rule applications, chart construction, context translations, and corrective revisions become equivalences when they carry the complements they would otherwise erase. Directed π PLN is the visible projection obtained by forgetting those inverse witnesses. Forgetting is then not necessarily destruction: it can be hidden residual transport or total holonomy. Double counting is not an unavoidable side effect of multiple proofs or charts: it is non-idempotent fusion after provenance, prior identity, or higher proof equivalence has been discarded.

The global probabilistic lesson is equally sharp. Local Bayesian coherence is valuable and fully retained. But a family of coherent local charts need not define one coherent world model. PGM–HoTT descent gives a principled criterion: use a global chart when the gluing type is geodesically contractible, retain guarded alternatives when it is multiply inhabited, and preserve the obstruction when it is empty.

The resulting architectural principle is:

$$\begin{aligned} \text{Geodesic } \pi\text{PLN should search an atlas-indexed directed higher proof space,} \\ \text{run ordinary Bayesian inference inside local charts,} \\ \text{and transport a reversible, idempotent evidence section across them.} \end{aligned} \tag{18.2}$$

This preserves the practical strengths of bidirectional least-effort reasoning while making context, provenance, revision, non-gluability, evidence reuse, and order-dependent curvature explicit, auditable, and separately measurable.

A Notation summary

Symbol	Meaning
$\mathcal{B}$	Directed higher proof base.
$\text{Core}(\mathcal{B})$	Maximal invertible core carrying PGM–HoTT identity.
$\text{Der}(x, y)$	Directed derivation space from x to y .
$\text{Ev}(x)$	Evidence fiber over inference state x .
T_p	Evidence transport along derivation p .
$p \simeq_{\mathcal{E}} q$	Strong evidence equivalence of parallel derivations.
ρ_{ctl}	Geodesic control mass on raw proof paths.
μ_{ev}	Unique-evidence measure on evidential components.
$\mathcal{D}_{\text{dc}}$	Duplicate-evidence functional.
$\mathcal{E}_{\text{mult}}$	Mode-weighted control/evidence multiplicity mismatch.
$R^{\mathcal{E}}$	Curvature of the evidence connection.
$H_{p,q}$	Relative holonomy between two reversible derivations.
q	Bregman dual residual or Dykstra correction.
Π	Forgetting projection from reversible microstate to visible PLN state.
$\mathcal{C}$	Category or restriction category of π PLN contexts.
$\mathcal{B}_C$	Directed higher proof space internal to context C .
$\mathcal{B}_{\text{tot}}$	Grothendieck construction of the context-indexed proof spaces.
Γ_C	Temporary local Bayesian or ν PLN chart in context C .
I_C, O_C	Evidence-to-probability import and prior-unwinding export maps.
$\text{Desc}_U(\Gamma_{\bullet})$	Homotopy-descent type of coherent global chart gluings.
$\mathcal{O}_{\text{desc}}$	Quantitative probabilistic chart-gluing obstruction.
$R_{\text{proof}}^{2,0}$	Within-context proof curvature.
$R_{\text{mixed}}^{1,1}$	Inference–translation mixed curvature.
$R_{\text{context}}^{0,2}$	Context and chart-transport curvature.
Rev	First-class evidence/context/ontology revision morphism.

B A compact axiom package for implementation

An implementation need not expose the full higher-categorical semantics. The exact discrete guarantees require the following package.

- A1.** Every evidence capsule has immutable primitive provenance or a sketch with a quantified metric error.
- A2.** Merge is associative, commutative, and idempotent on provenance identity.
- A3.** Pure inference does not mint new primitive evidence tokens.
- A4.** Rule transports preserve merge and are nonexpansive in the chosen evidence metric.

- A5.** Proof rewrites carry local evidence-coherence defects.
- A6.** Candidate rule pairs are connected in a sparse confusion graph when they share causal/evidential support.
- A7.** Reversible mode retains the data needed to reconstruct every visible rule input, including dual residuals for projection-like steps.
- A8.** Control mass and evidence mass are stored and reported separately.
- A9.** Every persistent evidence packet is indexed by a context and carries its assumption and ontology fingerprints.
- A10.** Local chart export removes chart priors and virtual counts before persistent storage.
- A11.** Cross-context fusion requires a covering context and explicit guarded translations.
- A12.** Chart, translation, and revision maps carry local coherence defects and reversible complements.
- A13.** The system records whether a selected chart family is uniquely gluable, multiply gluable, or non-gluable in the current model class.
- A14.** Revision is represented at packet level so quarantine/retraction/relabeling remain distinct from counterevidence.

Under A1–A4, exact or approximate de-duplication holds. Under A4–A6, the single-context proof-surface bound holds. A7 makes rule-loop holonomy executable, and A8 prevents a scheduler from silently turning proof multiplicity into confidence. A9–A14 provide the atlas-level typing, prior-unwinding, translation, coherence, descent, and revision conditions required by the π PLN architecture.

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